MILITARY SPECIFICATION

TEST SET RADAR AM/UPM-137

1. SCOPE

1.1 This specification covers Modular IFF Test Set AN/UPM-137. The test set is to be designed to calibrate and test Interrogators, Transponders, Video Coders-Decoders, and Radar Video Distribution Systems.

1.1.1 The test set will employ a flexible unitized plug-in design permitting convenient expansion of its functional capabilities.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-17/30	- Cable, Radio Frequency, Coaxial, RG-62A/U
MIL-C-17/73	- Cable, Radio Frequency, Coaxial, RG-212/U
MIL-C-17/84	- Cable, Radio Frequency, Coaxial, RG-223/U
MIL-S-901	- Shock Test; H.I. (High-Impact); Shipboard Machinery Equipment and Systems, Requirements for
MIL-Q-9858	- Quality Program Requirements
MIL-C-5015	- Connectors, Electric, AN Type, General Specification for.
MIL-C-5541	- Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-C-3767/12	- Connector, Plug, Electrical Power UP 131M
MIL-P-15024	- Plates, Identification - Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment
MIL-E-16400	- Electronic Equipment, Naval Ship and Shore: General Specification
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of
MIL-A-27434/3	- Adapter, Connector, Coaxial, Radio Frequency Between Series, Type UG 635/U
MIL-A-27434/4	- Adapter, Connector, Coaxial, Radio Frequency, Between Series, Type UG-636A/U
MIL-A-27434/37	- Adapter, Connector, Coaxial, Radio Frequency, Between Series, Type UG-201A/U
MIL-C-39012/6	- Connectors, Coaxial, Radio Frequency, (Series C Plugs, Class II)
MIL-C-39012/12	- Connectors, Coaxial, Radio Frequency, (Series C Receptacle, Flange Mounted, Rear Mounted, Class II)
MIL-C-39012/16	- Connectors, Coaxial, Radio Frequency, (Series BNC (Cabled) - Plug, Male, Class II)
MIL-C-39012/19	- Connectors, Coaxial, Radio Frequency, (Series BNC - Receptacles - Jam Nut Mounted, with Coaxial Cable Clamps-Class II)
MIL-C-39012/21	- Connectors, Coaxial, Radio Frequency (Series BNC (Uncabled)- Receptables, Female, Jam Nut Mounted, Class II)
MIL-C-39012/22	- Connectors, Coaxial, Radio Frequency, (Series BNC Receptacle, Flange Mounted -Class II)
MIL-C-39019	- Circuit Breakers, Magnetic, Low Power, Sealed, Trip Free, General Specification for

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STANDARDS

MILITARY

MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-167	- Mechanical Vibrations of Shipboard Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-681	- Identification Coding and Application of Hookup Wire
MIL-STD-756	- Reliability Prediction
MIL-STD-781	- Reliability Tests: Exponential Distribution
MIL-STD-785	- Reliability Program for Systems and Equipments Development and Production
MIL-STD-810	- Environmental Test Methods
MS 17321	- Meter, Time Totalizing 28 Volt, DC, Miniature, Digital
MS 35173	- Adapter, Connector, Series BNC, UG 274B/U

HANDBOOKS

MILITARY

MIL-HDBK-217	- Reliability Stress and Failure Rate Data for Electronic Equipment
DOD AIMS Specification 65-1000	- Technical Standards for the ATCRBS/IFF/Mark XII System (AIMS)

PUBLICATIONS

MILITARY

NAVSHIPS 94501	- Reliability Design Handbook
NAVSHIPS 0967-146-0010	- Dictionary of Standard Terminal Designations for Electronic Equipment

DRAWINGS

MILITARY

REB 49224	- UG-60E/U Jack for Use with RG-8/U and RG-9/U Cables
REB 49227	- UG-59E/U Plug for Use with RG-8/U and RG-9/U Cables
REB 49228	- UG-61E/U Connector Receptacle for Use with RG-8/U and RG-9/U Cables
REC 49438	- UG-273/U Adapter UHF Receptacle to BNC Plug
REB 49276	- UG-309/U Adapter, Series HN to Series BNC

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extend specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U.S. National Standards for Common System Component Characteristics for the IFF Mark X (SIF) Air Traffic Control Radar Beacon Systems SIF/ATCRBS

(Application for copies should be addressed to Federal Aviation Agency, Washington, D.C. 20590).

American Society of Heating and Ventilating Engineers Standards Code for Testing and Rating Filters

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(Application for copies should be addressed to the American Society of Heating and Ventilating Engineers, Inc., United Engineering Center, 345 East 47th Street, New York, New York 10017.)

3. REQUIREMENTS

3.1 Definitions. The definitions and descriptions of the Technical Standards for the ATCRBS/IFF/Mark XII Systems (AIMS), DOD AIMS 65-1000, apply unless otherwise specified herein.

3.1.1 Pulse shape. Pulse shape parameters shall conform to Figure 1 to aid in clarifying the definitions. "Emax" is the maximum voltage amplitude of the pulse as measured from the zero axis. Points B and C on the pulse envelope are at the 90 percent Emax level. Points A and D on the pulse envelope are at the 10 percent Emax level.

3.1.2 Pulse rise time. The rise time of a pulse is the time required for the leading edge of the pulse to rise from 10 to 90 percent of the maximum pulse amplitude. T_R, as shown in Figure 1, is the rise time and is the time required for the voltage to rise between points A and B. Unless otherwise specified herein, pulse rise time shall be between 0.05 and 0.1 microsecond (μsec).

3.1.3 Pulse decay time. The decay time of a pulse is the time required for the trailing edge of the pulse to decay from 90 to 10 percent of the maximum pulse amplitude. T_D, as shown in Figure 1, is the decay time and is the time required for the voltage to decay between points C and D. Unless otherwise specified herein, pulse decay time shall be between 0.05 and 0.20 μsec.

3.1.4 Pulse top. The pulse top shall be essentially flat. The instantaneous amplitude of the pulse between the 90 percent Emax points on the leading and trailing edges shall not fall below the 90 percent Emax level. On Figure 1, the amplitude of any point between B and C shall not fall below the level of the line through B and C.

3.1.5 Pulse duration. The pulse duration is the time between the 50 percent maximum voltage amplitude points on the leading and trailing edges. T_X, as shown in Figure 1, is the duration time.

3.1.6 Delay or spacing. Delay or spacing is the time between 50 percent maximum voltage amplitude points on the leading edges of the pulses concerned, except where otherwise specified herein.

3.1.7 Negative overshoot. The negative overshoot of a pulse during decay shall be less than 5 percent of Emax, the maximum pulse amplitude.

3.2 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.2).

3.3 For all nonstandard parts documentation must be submitted to the command or agency concerned in accordance with MIL-E-16400 for approval of the part prior to use in the equipment.

3.4 General requirements. The equipment shall meet the temperature requirements of class 4 of MIL-E-16400, except as hereinafter specified.

3.4.1 Reliability. The test set shall have a mean-time-between-failure (θ_0) of 300 hours, where θ_0 is defined by MIL-STD-781. The reliability goal for the test set shall be 900 hours MTBF.

3.4.1.1 Reliability program. The reliability requirements shall be accomplished through a reliability program plan in accordance with MIL-STD-785. The reliability program plan shall be submitted to the command or agency concerned for approval (see 6.1.2) after award of contract.

3.4.1.2 Reliability design guides. The contractor shall become familiar with publication NAVSHIPS-94501 Reliability Design Handbook and shall make maximum use of the design guides therein.

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3.4.1.3 Reliability prediction. A design reliability prediction shall be made using the methods contained in design prediction procedure paragraph of MIL-STD-756 and MIL-HDBK-217 or any alternate method approved by the command or agency concerned. The reliability mathematical model developed for the reliability prediction shall be submitted with the reliability prediction report. The source and applicability of all failure data used for the reliability prediction shall be documented. Failure data obtained from sources other than MIL-HDBK-217 is subject to the approval of the command or agency concerned. The final reliability prediction shall be submitted to the command or agency concerned before the initiation of first article testing or as specified in the contract or order.

3.4.1.4 Reliability demonstration. The reliability of the test set shall be demonstrated in accordance with the requirements of MIL-STD-781 and 4.5.4 of this specification.

3.5 Environmental conditions.

3.5.1 Temperature.

Operating	0°C to 50°C
Nonoperating	-62°C to +75°C

3.5.2 Altitude.

Operating	0 to 10,000 feet above sea level
Nonoperating	50.000 feet above sea level maximum

3.5.3 Vibration. The test set shall meet the requirements of Specification MIL-STD-167, Type I.

Frequency Range Hertz (Hz)	Table Amplitude (Peak) inch
5 to 15	0.030 ± 0.006
16 to 25	0.020 ± 0.004
26 to 33	0.010 ± 0.002

3.5.4 Shock. The test set shall meet the requirements for Grade A, Class I, Type A, of MIL-S-901.

3.5.5 Humidity. The test set shall be capable of operation at relative humidities ranging up to 95 percent for both continuous and intermittent periods, including conditions wherein condensation takes place in the form of water and frost.

3.5.6 Salt Spray. The salt spray shall be in accordance with method 811 of Federal Standard No. 151. Sample specimens of the top and bottom covers of the modules shall be subjected to 48 hours salt spray test. A corner section of the case representing the critical structure, shall be subjected to the 200 hours duration test. The individual modules and case shall not be subjected to the salt spray test.

3.5.7 Enclosure. The equipment shall be drip proof (15 degrees) in accordance with MIL-STD-108.

3.5.8 Inclination. The test set shall meet the inclination requirements of MIL-E-16400.

3.6 Electrical Design. The nominal supply voltage and frequency shall be 115 volts and both 60 and 400 Hz.

3.6.1 Overload protection shall be provided. The type of protection and type of indicator shall be subject to the approval of the command or agency concerned.

3.6.2 Maintenance. Test joints and test features are required and shall be subject to the approval of the command or agency concerned.

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3.6.3 Controls, indicators, and panel layout.

3.6.3.1 The design, location, and operation of all controls shall be subject to the approval of the command or agency concerned.

3.6.3.2 Control knobs. In the interest of conserving panel space, and with approval of the command or agency concerned, dual concentric control knobs may be employed to control related functions.

3.6.3.3 Front panel illumination shall not be required.

3.6.4 Rivets. Rivets shall not be used as a means of attaching potentially replaceable parts to any other part of the equipment.

3.6.5 Accessibility. All parts of the equipment shall be accessible for complete maintenance at the equipment's permanent installation. The location of all adjustments (electrical or mechanical) shall be such that maximum safety to the equipment and operating personnel is insured.

3.6.6 Instruction plates are not required, but required technical information shall be in the technical manual.

3.6.7 This equipment shall employ solid state and microelectronic techniques to the maximum extent practical.

3.6.8 Internal temperature. During the reliability tests, the average internal ambient temperature within the equipment shall not exceed 25°C beyond the external ambient under high line conditions.

3.6.9 Transients. Provisions shall be made to prevent damage to the CRT or other parts when the equipment is turned on or off.

3.6.10 Power consumption. The manufacturer shall employ all methods feasible in the process or manufacture, commensurate with good engineering practice and the requirements of this specification to keep the power consumption of the equipment to a minimum.

3.6.11 Size. The equipment shall not be greater than 32-1/4 inches wide by 20-3/8 inches deep by 23-1/4 inches high. Mounting pads may extend beyond the rear by 2.5 inches, but shall be capable of being rotated beneath the main enclosure.

3.6.12 Ventilation. All apertures, louvers, and other openings shall be screened or attenuated where there is a possibility of radio-frequency leakage.

3.6.12.1 Air circulation. Sufficient internal air circulation and heat conduction paths shall be provided to all parts of the equipment to prevent the failure of parts due to excessive heating.

3.6.12.2 Space heaters. Space heaters shall be employed, if required, in units in order to insure operation under the humidity conditions specified in MIL-E-16400.

3.6.12.3 The temperature of all operating panels shall be less than 43°C when the ambient temperature in the space is 25°C.

3.6.13 In the event that the equipment becomes unbalanced when the drawers are opened, warning plates shall be provided.

3.6.14 The standard colors to be used in chassis wiring shall be as specified in MIL-STD-681.

3.6.15 Color coding of tube sockets will not be required.

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3.6.16 Terminal designations. Standard terminal numbers shall be subjected to the approval of the procuring agency and may be in accordance with NAVSHIPS 0967-146-0010. Terminals which are standardized include the following:

a. Terminals of this equipment used for connecting units of this equipment to units of other equipment.
b. Terminals for primary power, trigger input, trigger output, video input, and video output used in interconnecting units of this equipment.

3.6.17 Transformers. All transformers shall be in accordance with MIL-E-16400 except as otherwise specified in 3.6.17.1 and 3.6.17.2.

3.6.17.1 Transformer marking. Power transformers shall be marked with winding and terminal designations stenciled on the case or printed on a plate attached to the case. The marking shall be easily read when the transformer is installed in the equipment.

3.6.17.2 All power transformers shall withstand 25 percent increase in nominal line voltage for at least 30 minutes. During the test the transformers shall be fully loaded.

3.6.18 Electromagnetic interference emission, susceptibility, and equipment and systems selfcompatibility. The test set shall operate within the electromagnetic interference emission, susceptibility, and equipment and systems selfcompatibility limits in accordance with class IC of MIL-STD-461, except that CE01 is not required.

3.6.18.1 The equipment shall be so designed as to be capable of accurately measuring the receiver sensitivity of IFF equipments to a level of -95 dBm minimum.

3.6.18.2 Radiation protection. Protection and shielding shall be provided to protect personnel from the harmful effects of X-rays and other radiation. The method to be used to secure this protection shall be subject to the approval of the command or agency concerned. No protection is necessary if the CRT is used with less than 10,000 volts, and if solid state rectification is used in high voltage lines.

3.6.19 Tools. In addition to the requirements of MIL-E-16400, no tools other than an ordinary screwdriver shall be required to gain access to the several chassis for servicing.

3.6.20 Circuit cards. The design and use of all cards shall be subject to the approvel of the command or agency concerned.

3.6.20.1 Circuit card keying. All plug-in circuit cards shall be uniquely keyed to prevent accidental insertion into the wrong circuit card connector. The keying shall also prevent inadvertant reversal of a circuit card whether directly plugged into the connector, or extended by means of the accessory extender board (see 3.7.6.8).

3.6.21 Modules. The test set shall consist of four independent plug-in modules. The design shall be such that any one module may be removed from the equipment case and operated on a bench for maintenance purposes. Each module shall contain all panel controls and signal input/output jacks peculiar to its function. Inter-module connections shall be through the rear of the unit to minimize cable patches between modules during operation to the maximum extent practicable.

3.6.22 Weight. The weight of any one plug-in module exclusive of the power supply (see 3.7) shall not exceed 30 pounds and the total weight of the test set shall not exceed 190 pounds.

3.6.23 Panel protection. Handles shall be provided on the front panel of each module to protect the front panel controls and indicators from damage and for use in handling the modules.

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3.6.24 Finish. Aluminum parts shall be chemically treated in accordance with Type I or II, Grade A or B, Class 2 of MIL-C-5541.

3.6.25 Identification plates. The test shall have an identification plate in accordance with MIL-P-15024.

3.6.26 Power supplies. All ac and dc voltages required by the test set (except the high voltage required by the CRT) shall be supplied by a common power supply located in the equipment case. The power ON-OFF switch, indicator lamp, and protective device shall be located in the front of the case.

3.6.27 IFF system compatibility. This equipment shall meet the U.S. National Standards for Common System Component Characteristics for the IFF Mark X (SIF)/Air Traffic Control Radar Beacon System SIF/ATCRBS.

3.7 Equipment composition. The Radar Test Set AN/UPM-137 shall consist of the following:

(a) Equipment Case as specified in 3.7.1.
(b) SIF Generator Module as specified in 3.7.2.
(c) Oscilloscope Module as specified in 3.7.3.
(d) Interrogator Module as specified in 3.7.4.
(e) RF Module as specified in 3.7.5.
(f) Accessories and accessory box as specified in 3.7.6.

3.7.1 Equipment case.

3.7.1.1 General. The case shall be constructed in accordance with MIL-E-16400 and shall be drip proof in accordance with MIL-STD-108. Carrying handles shall be employed where required; their design and placement shall be approved by the command or agency concerned.

3.7.1.2 Case, electrical equipment. This case shall accept the four test set modules and accessories as specified in 3.7.

3.7.1.3 Power input. A connector shall be provided on the rear of the equipment and shall be capable of accepting the input power. The connector shall be a type MS3102A14A-5P in accordance with MIL-C-5015 or equivalent.

3.7.1.4 Protective cover. A removable cover shall be provided which will fasten to the front of the unit to protect the controls, jacks and indicators. Cover material shall be similar to the material used for the construction of the equipment case. The intent of the cover is for protection during shipment or storage only and it shall be removed during any operation or testing of the equipment. Design of the cover and fastening method shall be subject to approval by the command or agency concerned.

3.7.1.5 Safety stops. Each plug-in module shall have associated with it at least one safety stop. The stop device shall prevent total removal of the module from the equipment case unless specifically disengaged by the operator. The stop device shall automatically reengage when the module is reinserted in the equipment case.

3.7.2 SIF generator module.

3.7.2.1 General design. Basic design of the SIF generator module shall permit removal of the unit from the equipment case and allow for operation separately from the rest of the test set for maintenance purposes. Figure 2 is the approved front panel layout of the SIF Module. Deviations from this configuration require prior approval by the command or agency concerned.

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3.7.2.2 Circuit Elements. The equipment functions shall be provided by eight functional areas:

(a) SIF Coder Unit No. 1 (SIF Code Generation).
(b) SIF Coder Unit No. 2 (Interleaved SIF Code Generation).
(c) Trigger and Delay Unit.
(d) Mode Generator Unit.
(e) Challenge/Tag-Reset and Mixed Video-Generator Unit.
(f) Mode 4 (Word A) Maintenance Word Generator Unit.
(g) Mode 4 (Word B) Maintenance Word Generator Unit.
(h) Suppressor Pulse Generator Unit.

3.7.2.3 Types of Pulses. The following types of pulses or groups of pulses are referred to in this module.

(a) IFF Interrogation:

Type	Number of Pulses	Width (μsec)
I	2 or 3 (variable spacing)	Less than 0.30 to at least 1.3

(b) Selective indentification feature (SIF):

Type	Number of Pulses	Width (μsec)
2	2 to 15	0.45 nominal 0.3 to 0.55 nominal variation, less than 0.3 to at least 1.0 extreme variation.

(c) Mode 4:

Type	Number of Pulses	Wide (μsec)
3 (interrogation)	Up to 37	Less than 0.2 to at least 0.7
4 (reply)	3	Less than 0.2 to at least 0.7

3.7.2.4 SIF Coder Unit No. 1. The SIF No. 1 coder shall be capable of generating pulse trains, as specified herein; containing none to a maximum of 13 information pulses plus two framing pulses. The pulse trains shall be generated at any repetition frequency between 10 and 11,000 pulses per second (pps), when the coder is triggered from either of the triggers specified in 3.7.2.6.1.2 or 3.7.2.6.3. The coder output shall be capable of variable pulse width and variable output level controlled as specified in 3.7.2.4.4 and 3.7.2.12.1 respectively.

3.7.2.4.1 Pulse spacing. All pulses shall be capable of internal adjustment about their ideal position to a tolerance determined by the marker accuracy specified in 3.7.3.5.5. After the pulses are set to this accuracy, the position of each pulse shall remain within plus or minus 0.05 μsec of the ideal pulse position throughout all environmental contitions. When measured under normal room ambient, and after a one-half hour warm-up period, the shift in pulse positions shall not exceed plus or minus 0.02 μsec from the ideal position.

3.7.2.4.2 Residual delay. The delay between the trigger input to the code generator and the start pulse shall be 3.0 ±0.5 μsec.

3.7.2.4.3 Substitute pulse facility. A single front panel control shall delete any selected pulse in the train but the start pulse, and substitute in its place a pulse of identical characteristics capable of continuously

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variable position. Visual means shall be provided to indicate which pulse of the train is being replaced by the substitute pulse. A separate front panel control shall vary the position of the substitute pulse to plus or minus 1.6 μsec of the position of the deleted pulse. The substitute pulse control shall be calibrated and detented at the "0" position. The "0" position shall maintain a tolerance of ±0.05 μsec throughout all environmental conditions. When measured under normal room ambient and after a one-half hour warm-up period, the shift in pulse position shall not exceed plus or minus 0.02 μsec. The above tolerances apply only to the substitute pulse and shall refer to the ideal position of the deleted pulse. The control shall be indexed at the ±0.5 μsec and ±1.0 μsec positions.

3.7.2.4.4 Pulse width. The width of each pulse in the train shall be simultaneously variable over a continuous range of less than 0.3 to greater than 1.0 μsec by means of a single front panel control. Nominal pulse width shall be 0.45 μsec.

3.7.2.4.5 Coder output. The amplitude variation of pulses in any train shall be less than 2 percent of the average maximum amplitude. All other variations in pulse shape shall be less than 5 percent.

3.7.2.4.6 Code number assignments. Facilities shall be provided for the generation of 12 information pulses. It is not intended to employ the midposition pulse between the framing pulses. The applicable 12 information pulses shall be divided into four groups of three pulses. Each pulse in a group shall be separated by 2.9 μsec and shall be serially numbered in order 1, 2, and 4. The position of the first pulse of each group from the first framing (start) pulse shall be as follows:

Group	Time from first framing pulse (μsec)
A	2.90
B	11.60
C	1.45
D	13.05

Code numbers 0000 through 7777 shall designate the appropriate code numbers. The code number shall be determined by adding the pulse numbers (1, 2 and 4) of the pulses present within each group. These sums shall then be combined in the order A, B, C, D to form a four digit serial number for the particular code. For example: Code 2435 consists of pulse 2 of group A, pulse 4 of group B, pulses 1 and 2 of group C, and pulses 1 and 4 of group D. The particular pulse train for Code 2435 will, therefore, have information pulses at the first, third, fourth, ninth, twelfth, and thirteenth positions.

3.7.2.4.7 Code Selector. A facility for selecting code numbers 0000 through 7777 shall be provided on the front panel.

3.7.2.4.8 The code selection facility and front panel arrangement shall be approved by the command or agency concerned. The requirements of 3.7.2.4.6 shall not exclude the ability to simultaneously generate any combination of 12 information pulses plus the two framing pulses.

3.7.2.4.9 Function Control. A front panel function control switch shall be provided to select the following types of reply codes:

(a) Emergency
(b) Identification of Position
(c) Normal
(d) "X" Pulse
(e) Variable Emergency

3.7.2.4.9.1 Emergency. A front panel switch shall be provided for the emergency facility. When this switch is in the "ON" position, any Mark X (SIF) pulse-coded reply train designated by the code selector switches shall be generated, followed by three pairs of bracket pulses. The spacing between the leading edges of the last bracket pulse of one train and the first bracket pulse of the following train shall be 4.35 + 0.1 μsec. The emergency function shall not affect Mode 4 operation.

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3.7.2.4.9.2 Identification of Position. When the function control switch is in this position, any Mark X (SIF) pulse train code designated by the Code Selector Switch is followed by the first bracket pulse only of the second train. The spacing between the first bracket pulse of the first train and the first bracket pulse of the following train shall be 24.65 ±0.1 μsec.

3.7.2.4.9.3 Normal. When the function control switch is in this position, the reply code is as designated by the Code Selector Switches.

3.7.2.4.9.4 "X" Pulse. When the function control switch is in this position, any normal type reply shall contain an additional pulse positioned 10.15 ±0.05 μsec from the first bracket (start) pulse.

3.7.2.4.9.5 Variable Emergency. When the function control switch is in this position, it shall be possible by means of a separate panel control to vary the interval between F2 of the preceding bracket pair and F1 of the next bracket pair over a range of ±1 μsec about the ideal spacing of 4.35 μsec. The variable spacing control shall cause all F2-F1 intervals to vary simultaneously. The variable emergency function shall not exclude the ability to vary the position of individual pulses in the initial code train utilizing the substitute pulse facility.

3.7.2.5 SIF Coder Unit No. 2. The equipment shall have a second SIF Code Generator which is completely independent of SIF Coder Unit No. 1 to permit generating interleaved trains.

3.7.2.5.1 Trigger. An independent trigger and delay shall allow continuous train to train displacement between coder No. 2 and coder No. 1 over the range of less than 0.5 to at least 30 μsec.

3.7.2.5.2 Functions. Selection of SIF Coder No. 2 by a front panel control shall allow any normal reply code to be generated and mixed with the selected outputs of coder No. 1. Code selection, pulse width adjustment and substitute pulse facility shall be identical to coder No. 1. However, coder No. 2 shall generate the normal code only, and shall not be required to generate Emergency, Identification of Position or "X" codes.

3.7.2.5.3 Outputs. SIF Coder No. 2 shall be available as separate video signals and as part of the mixed video.

(a) When available as a mixed video, the amplitude shall be as specified in 3.7.2.12.1.
(b) When available as separate video, the amplitude shall be +5V ±10 percent across 75 ohms.

The separate SIF Coder No. 2 video output shall be made available at a front panel jack type UG-625B/U in accordance with MIL-C-39012/21 or equivalent.

3.7.2.5.4. SIF Target Gate. A jack shall be provided on the front panel of the unit to accept a target gate such as is generated by the Interrogator Module (see 3.7.4.6.7). When the gate is connected to the jack, SIF Coder output shall occur only during the gating period, and shall be inhibited at all other times. The inhibiting signal will have the following characteristics.

Gate Off	0 v ±0.5 v
Gate On	2 to 5 volts positive

The front panel jack shall be a type UG-625B/U in accordance with MIL-C-39012/21 or equivalent.

3.7.2.6 Trigger and Delay Unit. This unit shall perform the following functions:

3.7.2.6.1 Triggers. This unit shall generate the following triggers which shall be separately available by means of UG-625B/U panel jacks in accordance with MIL-C-39012/21 or equivalent located on the front panel. The unit will operate self-synchronous at a variable rate, or in synchronization with an external positive or negative trigger (see 3.7.2.6.3). When triggered internally, the repetition rate shall be continuously variable over a linear range of 10 to 11,000 pps, for both the "0" trigger and delay trigger by means of a front panel control. For Mode 4 operation, the internal trigger PRF shall be continuously variable over a linear range of 5 to at least 5500 pps.

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3.7.2.6.1.1 "0" (Zero) Delay Trigger. The output of the internal synchronization trigger generator is defined as the "0" delay reference. When the equipment is externally synchronized, the output of the "0" trigger jack shall be delayed by not more than 0.25 μsec with respect to the external trigger.

3.7.2.6.1.2 Delay Trigger. Provisions shall be made for delaying the internal and output triggers as well as the external trigger from 0.4 to 4400 μsec. The delay shall be continuously variable and shall be selected by means of a front panel control, or controls, which give an approximately linear relation between control rotation and delay interval. The sensitivity of the control(s) shall be such that a specific delay can be set to within 0.01 μsec ±0.01 percent of the total delay. The control(s) shall have the following ranges:

(a) No Delay - The output trigger coincident with the "0" trigger within 0.25 μsec.
(b) Less than 1 to 4.4 μsec continuously variable by means of a multiturn control and indicator.
(c) Additional ranges of 4.0 to 44, 40 to 440 and 400 to 4400 μsec. It shall be possible to set the delay to a specifie value within this range and to a tolerance no greater than the jitter. Fine and coarse controls shall be used. Jitter with respect to the internal or external initiating trigger shall be less than plus or minus 0.01 μsec ±0.01 percent of the delay time.
(d) The trigger pulse shall have the characteristics as specified in 3.7.2.6.5. Jitter requirements of 3.7.2.6.1.2(c) apply.

3.7.2.6.2 By means of internal wiring, the trigger specified in 3.7.2.6.1.2 shall be made available to the coder units specified in 3.7.2.4 and 3.7.2.5. By means of internal cabling, the triggers specified in 3.7.2.6.1 shall also be made available to the Oscilloscope Module, and the trigger specified in 3.7.2.6.1.2 shall be made available to the Interrogator Module.

3.7.2.6.3 External Trigger (See 3.7.2.6.1). The equipment shall accept an external trigger, or input, having the following characteristics:

(a) Positive or negative polarity.
(b) Any repetition rate between 15 and 4100 pps.
(c) Any pulse duration between 0.3 and 25 μsec.
(d) A peak voltage of from 5 to 50 volts measured across a 75 ohm ±5 percent terminating resistor. Nominal 20 ±5 volts.
(e) Rise time less than 0.5 μsec per volt.

3.7.2.6.4 The external trigger shall be applied through a UG-625B/U front panel jack in accordance with MIL-C-39012/21 or equivalent. The equipment shall not be damaged by pulses of the following characteristics: 200 volt peak amplitude, 250 μsec duration, and having repetition rates up to 2,000 pps; or by 500 volts direct current (dc) applied to the external trigger jack. The input impedance shall be not less than 100,000 ohms in parallel with 50 micormicrofarads (μμf).

3.7.2.6.5 Trigger Output Pulse. The trigger output pulse shall be 20 ±5 volts positive amplitude when working into 75 ohms in parallel with 1,100 μμf. The pulse shall have a duration of not less than 1.0 μsec nor more than the rise time of the pulse measured from the 10 percent maximum amplitude point to the 90 percent level shall be not greater than 0.1 μsec. The decay time measured from the 90 percent level to the 10 percent maximum amplitude level shall be not greater than 0.2 μsec.

3.7.2.7 Mode Generator Unit.

3.7.2.7.1 Challenge Mode. The unit shall be capable of generating the control functions for any one or more of five challenge modes. The modes shall be mode 1, mode 2, mode 3/A, mode C, and mode 4.

3.7.2.7.2 Mode Selection. By means of a separate switch for each mode on the front panel, it shall be posible to select and interlace two or more modes. The selected mode control functions shall interlace in order. Only the selected modes shall appear. Unselected modes shall be skipped and the next mode control function in order shall be generated without timing gaps. The mode interlace capability shall also permit the capability of generating a single continuous mode control function.

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3.7.2.7.3 Mode repeat. A control located on the front panel shall perform the function of mode repeat selection. It shall be possible to repeat all selected mode control functions for a selected number of PRF cycles. Mode repeat multipliers shall be 1, 2, 3, 4, 8 and 16 times. For example, if modes 1 and 3 are selected and mode repeat multiplier is 4, the mode control functions generated by subsequent PRF cycles will be 1111333311113333 etc.

3.7.2.7.4 Challenge tag-reset and mixed video unit. Under control of the mode selection and mode repeat controls, the unit shall perform the following functions.

3.7.2.7.4.1 Challenge - Mode 1. Type 1 pulses having spacings variable from less than 1.0 to greater than 5.0 μsec between leading edges. The pulse spacing shall be variable within specified limits by varying the position of the first pulse with reference to the leading edge of the second pulse. A detent shall be provided at the 3 ±0.05 μsec position.

3.7.2.7.4.2 Challenge - Mode 2. Type 1 pulses haveing spacings variable from less than 3.0 to greater than 7.0 μsec between leading edges. The pulse spacing shall be variable within specified limits by varying the position of the first pulse with reference to the leading edge of the second pulse. A detent shall be provided at the 5 ±0.05 μsec position.

3.7.2.7.4.3 Challenge - Mode 3/A. Type 1 pulses having spacing variable from less than 6.0 to greater than 10.0 μsec between leading edges. The pulse spacing shall be variable within specified limits by varying the position of the first pulse with reference to the leading edge of the second pulse. A detent shall be provided at the 8±0.05 μsec position.

3.7.2.7.4.4 Challenge - Mode C. Type 1 pulses having spacings variable from 19 to 23 μsec between leading edges. The pulse spacings shall be variable within the specified limits by varying the position of the first pulse with reference to the leading edge of the second pulse. A detent shall be provided at the 21.0±0.05 μsec position.

3.7.2.7.4.5 Sync Mode 4. Four Type 3 pulses spaced at 2.0 ±0.05 μsec intervals The first three pulses shall be variable in turn at least ±2 μsec about their ideal position. A detent shall be provided at the nominal position.

3.7.2.7.5 Pulse Width Control. A control located on the front panel shall vary the pulse width of all pulses generated by the coder from less than 0.3 to at least 1.3 μsec for modes 1, 2, 3/A, and C and from less than 0.2 to at least 0.7 μsec for mode 4. During interlaced operation, the adjustment range shall automatically switch to the appropriate range for the mode being generated.

3.7.2.7.6 ISLS Control Pulse. Provisions shall be made for the generation of an ISLS control pulse. The ISLS (P2) pulse shall be triggered 2 ±0.05 μsec following the interrogation code P1 pulse, providing the specific P2 functional characteristics. For mode 4, the ISLS pulse shall follow the first sync pulse 8 ±0.07 μsec.

3.7.2.7.6.1 ISLS pulse spacing control. Provisions shall be made to vary the ISLS control pulse spacing over the range from 1.75 to 2.25 μsec for Mode 1 operation and 1.6 to 2.5 μsec for Modes 2, 3/A, C and 4 operation. The pulse spacing shall be referenced from the first interrogation pulse (or the fourth sync pulse for mode 4). An indice and detent shall be provided for setting the pulse spacing to 2.0 ±0.05 μsec.

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3.7.2.7.6.2 ISLS pulse width control. This shall provide continuous control of pulse width over a range of less than 0.3 μsec to greater than 1.3 μsec for modes 1, 2, 3/A and C, and from less than 0.2 to at least 0.7 μsec for mode 4. An indice shall be provided for setting the control to 0.8 ±0.05 μsec for modes 1, 2, 3/A, and C and 0.5 ±0.05 μsec for mode 4. During interlaced operation, the adjustment range shall automatically switch to the appropriate range for the mode being generated.

3.7.2.7.6.3 ISLS amplitude. The amplitude of the ISLS pulse measured at the coder out jack shall be 20 ±5 V into 75 ohm ±5 percent. The rise and decay times of the output pulse shall be as follows:

T_r 0.05 to 0.1 μsec

T_D 0.05 to 0.15 μsec

3.7.2.7.6.4 ISLS output jack. A panel jack type UG-625B/U in accordance with MIL-C-39012/21 or equivalent shall be provided on the front panel of the module to serve as the output from the ISLS generator.

3.7.2.7.6.5 Mixed ISLS control. A switch shall be provided on the front panel of the module to permit mixing of the ISLS pulse with the challenge pulses and the pulse trains of 3.7.2.8 and 3.7.2.9. When the switch is in the "SLS IN" position, the ISLS pulse shall be mixed with the challenge/tag output at the challenge/tag output jack. When the switch is in the "OFF" position, the ISLS pulse shall not mix with the challenge/tag output but shall nevertheless appear at the SLS OUT jack. The switch shall have a spring loaded third position labeled "DISPARITY" such that when engaged the pulse in the 30 μsec position of both A and B maintenance words (see 3.7.2.8 and 3.7.2.9) shall be deleted.

3.7.2.7.7 Mixed Level Control. The challenge/tag level control shall permit simultaneous control of the mixed tag video out and ISLS signals.

3.7.2.8 Mode 4 (Word A) Maintenance Word Generator Unit. The Mode 4 (Word A) Maintenance Word Generator shall be capable of generating the pulse train specified herein. The generator shall be internally triggered by the ZERO trigger. The external trigger circuitry shall be disabled when Mode 4 maintenance word generator is operated.

3.7.2.8.1 Pulse Spacing. A pulse train shall be generated consisting of 28 pulses spaced 0, 2, 4, 6, 10, 13, 15, 17, 20, 24, 26, 30, 33, 35, 38, 41, 43, 45, 48, 50, 52, 54, 58, 61, 63, 66, 68 and 71 μsec with respect to the leading edge of the first pulse. No pulse of the train shall vary more than ±0.07 μsec from its stated position. All pulses shall be 0.5 ±0.05 μsec wide.

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3.7.2.9 Mode 4 (Word B) maintenance word generator unit. The Mode 4 (Word B) Maintenace Word Generator shall be capable of generating the pulse train specified herein. The generator shall be internally triggered by the Zero trigger. The external trigger circuitry shall be disabled when the Mode 4 maintenance word generator is operated.

3.7.2.9.1 Pulse spacing. A pulse train shall be generated consisting of 28 pulses spaced 0, 2, 4, 6, 10, 12, 15, 17, 19, 22, 25, 28, 30, 32, 35, 37, 40, 44, 48, 51, 53, 57, 60, 62, 64, 66, and 70 μsec with respect to the leading edge of the first pulse. No pulse of the pulse train shall vary more than ± 0.07 μsec from its stated position. All pulses shall be 0.5 ± 0.05 μsec wide.

3.7.2.10 Suppressor pulse generator unit. The Suppressor Pulse Generator Unit shall be capable of generating the suppressor pulse specified herein.

3.7.2.10.1 Suppressor pulse. The suppressor pulse output shall be provided at essentially "0" time. The suppressor pulse shall be continuously variable over the specified range by means of a front panel control. The suppressor pulse shall be triggered by the "0" triggers.

(a) Amplitude. The amplitude shall be greater than 20 volts positive when working into a load of 500 ohms in parallel with 175 μμf. The amplitude shall not be less than 15 volts when working into a load of 75 ohms in parallel with 1,100 μμf.
(b) Rise time. The rise time shall be less than 0.4 μsec up to the 15 volt level.
(c) Decay time. The decay time shall be less than 0.4 μsec between the 15 volt level and 10 percent of peak amplitude.
(d) Duration. The pulse duration shall be variable from 2.0 to 220 μsec.
(e) Duty cycle. The duty cycle of the suppressor pulse need not exceed 20 percent.

3.7.2.11 Tagged video system.

3.7.2.11.1 Tagged video. In this equipment, the tagged video system employs interrogation pulses as the identification tag, normal reply video, and a characteristic reset signal.

3.7.2.11.2 Identification tags. The identification tags shall consist of the following:

(a) Mode 1. 2 pulses 0.8 ± 0.1 μsec wide spaced 3 ± 0.1 μsec.
(b) Mode 2. 2 pulses 0.8 ± 0.1 μsec wide spaced 5 ± 0.1 μsec.
(c) Mode 3/A. 2 pulses 0.8 ± 0.1 μsec wide spaced 8 ± 0.1 μsec.
(d) Mode C. 2 pulses 0.8 ± 0.1 μsec wide spaced 21 ± 0.1 μsec.
(e) Mode 4. 4 pulses 0.5 ± 0.1 μsec wide spaced 2 ± 0.1 μsec.
(f) Sidelobe suppression pulses shall be available in all five modes. For Modes 1, 2, 3/A and C, these pulses shall occur 2 ± 0.05 μsec after the first pulse and be 0.8 ± 0.1 μsec wide. For Mode 4, the pulse shall occur 8 ± 0.07 μsec after the first pulse and be 0.5 ± 0.1 μsec wide.
(g) Provisions shall be made to allow any of the pulses in (a) through (e) to be varied about its normal position at least ± 2.0 μsec (see 3.7.2.11.5) and provisions shall be made to allow the pulse in (f) to be varied about its normal position from at least -0.4 to at least +0.5 μsec. (See 3.7.2.7.6.1)

3.7.2.11.3 Reset signal. The reset signal shall consist of three pulses. The first pulse shall be 8 ± 0.2 μsec wide, the second pulse 2.0 ± 0.25 μsec wide shall occur 10 ± 0.1 μsec from the half voltage point of the first pulse. The third pulse shall be 4.0 ± 0.25 μsec and shall occur 15 ± 0.1 μsec from the half voltage point of the first pulse. It shall be possible to move any of the reset signal pulses ± 2.0 μsec about its ideal position.

3.7.2.11.4 Tagged video signals. The tagged video signals shall be available as separate video signals and as part of the mixed video, including SIF trains, and capable of modulating the signal generators.

(a) The mode identification tags shall be available separately.

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(b) The reset signals shall be available separately.
(c) The mixed video shall be available separately.

3.7.2.11.5 Challenge/tag-reset pulse positioning. A control, with detents in accordance with 3.7.2.7.4 shall be provided on the front panel for varying the position of the challenge/tag or reset pulses, about their ideal positions. The time reference challenge/tag pulses shall be the last pulse for modes 1, 2, 3/A, and C or the 4th pulse for mode 4. The time reference pulses for all modes shall occur in time coincidence. For the reset signals, the first pulse shall be considered the time reference pulse. Provision shall be made to allow any of the pulses, except the time reference pulses to be moved.

3.7.2.11.6 Variable pulse selection. A seven position control shall be provided to select the desired challenge/tag or reset pulse to be moved. The control functions shall be as follows:

Off	The pulse position variation control is disabled.
P1	Selects the P1 pulses of modes 1, 2, 3/A and C for position variation.
M4-1	Selects the first mode 4 sync pulse for position variation.
M4-2	Selects the second mode 4 sync pulse for position variation.
M4-3	Selects the third mode 4 sync pulse for position variation.
R2	Selects the second reset pulse for position variation.
R3	Selects the third reset pulse for position variation.

3.7.2.11.7 Mixed video control. A control shall be provided on the front panel to select the mixed video functions consisting of the following:

(a)	SIF 1	SIF 1 output shall appear alone. The train shall be triggered by the delay trigger.
(b)	SIF 1 and 2	Both SIF 1 and SIF 2 signals shall appear as mixed video. Timing of SIF 1 shall be the same as the previous function, (a).
(c)	Tag-Reset	The tag and reset signals alone shall appear as mixed video. The tags shall be triggered by the delay trigger. The reset shall follow the tag P3 pulse by the amount of the reset delay (3.7.2.11.8).
(d)	Tag-SIF-Reset	The tag, SIF 1 and reset signals shall appear as mixed video. The tags shall be triggered by the delay trigger. The SIF 1 shall follow the tag P3 pulse by the amount of the SIF delay (3.7.2.11.8). The reset shall follow the tag P3 by the amount of the reset delay.
(e)	All SIF	The tag, SIF 1, SIF 2 and reset signals shall all appear as mixed video. The timing shall be the same as the previous function, (d).
(f)	Mode 4 (Word A)	Mode 4 (Word A) shall appear alone.
(g)	Mode 4 (Word B)	Mode 4 (Word B) shall appear alone.

3.7.2.11.8 SIF and reset delays. The SIF and reset delays shall be separate but identical delay trigger circuits. The delays shall be continuously adjustable from 30 to 3000 μsec. A multiturn counting dial on the front panel shall provide delay readout of ± 5 percent accuracy. Jitter requirements of 3.7.2.6.1.2(c) apply.

3.7.2.12 Video outputs.

3.7.2.12.1 Mixed video outputs. Mixed video outputs shall have the following nominal characteristics with a 75 ohm termination.

(a) Variable; less than 0.1 volt to 5 volts, Tr 0.05 -0.1 μsec, Td 0.05 -0.15 μsec.
(b) High Level; 20 ± 5 volts minimum, T_r 0.05-0.1 μsec, T_d 0.05-0.15 μsec.

3.7.2.12.2 Mixed video output jacks. Two type UG-625B/U jacks in accordance with MIL-C-39012/21 or equivalent, shall be used for the front panel outputs of the coder.

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3.7.2.12.3 Challenge/tag output. A separate challenge/tag video output shall be provided.

3.7.2.12.3.1 By the operation of a front panel control it shall be possible to set the output at the challenge/tag out jack to any amplitude from less than 0.1 volt to the maximum output.

3.7.2.12.3.2 Pulse amplitude. The amplitude of the challenge/tag measured at the challenge/tag jack with the challenge/tag level control set for maximum output, shall be at least 15 volts into 75 ohms ± 5 percent. With the challenge/tag level control set for maximum output and terminated into 75 ohms ± 5 percent, the rise and decay times of the output shall be as follows:

T_r 0.05 to 0.1 μsec
T_d 0.05 to 0.15 μsec

3.7.2.12.3.3 Challenge/tag out jack. A panel jack type UG-625B/U in accordance with MIL-C-39012/21 or equivalent shall be provided on the front panel of the module to serve as the output from the coder.

3.7.2.13 Reset output. Separate reset video outputs shall be provided as follows:

(a) As a reset video signal the amplitude shall be adjustable from less than 0.1 volt to 5 volts across 75 ohms.
(b) As a reset trigger signal, the amplitude shall be 20 ± 5 V across 75 ohms.

3.7.2.13.1 Reset out jacks. Type UG-625B/U jacks in accordance with MIL-C-39012/21 or equivalent shall be used for front panel reset video and reset trigger outputs.

3.7.3 Oscilloscope module.

3.7.3.1 General design. Basic design of the oscilloscope module shall permit removal of the unit from the equipment case and allow for operation separately from the rest of the test set for maintenance purposes. The approved front panel layout for the oscilloscope module is shown in Figure 3. Deviations from this configuration require prior approval by the command or agency concerned.

3.7.3.1.1 Basic functions. The oscilloscope module shall contain a full dual trace oscilloscope. Dual display shall be obtained by alternating sweeps of channel No. 1 and channel No. 2. Channel No. 1 shall display video inputs which may be algebraically summed with frequency markers inserted at the mark input on the front panel. Channel No. 2 shall display video inputs or internal crystal controlled markers. Both channels contain identical attenuators, vertical amplifiers, video input jacks, jack load switches, and AC/DC switches.

3.7.3.1.2 Circuit elements. The equipment functions shall be provided by six functional areas:

(a) Sweep trigger delay unit.
(b) Sweep unit.
(c) Marker unit.
(d) Display unit.
(e) Vertical unit.
(f) Calibration.

3.7.3.2 Sweep trigger delay. The sweep delay function shall be controlled by a combination of two separate front panel control knobs. All sweep delays shall overlap and provide the following functions which give an approximately linear relation between control rotation and delay interval. The sensitivity of the control(s) shall be such that a specific delay can be set for ± 0.1 μsec ± 0.01 percent of the total delay

(a) No delay sweep will be enabled within 0.25 μsec of the "0" trigger.
(b) Less than 1 to 4.4 μsec continuously variable delay.
(c) 4.0 to 44 μsec continuously variable delay.
(d) 40 to 440 μsec continuously variable delay.

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(e) 400 to 4400 μsec continuously variable delay.
(f) Jitter and control resolution shall be as specified in 3.7.2.6.1.2.

3.7.3.3 Synchronization selector. A control shall be located on the front panel of the module and perform the following functions:

(a) Select external synchronization, + or - and, ac or dc.
(b) Select internal synchronization, + or -.
(c) Select crystal marker synchronization. (On internal synchronization only.)

3.7.3.4 Channel selection. A front panel switch shall permit selection of display as follows:

(a) Channel No. 1 alone.
(b) Channel No. 2 alone.
(c) Alternate display of both channels on alternate triggers.

3.7.3.5 Crystal controlled markers. Internal crystal controlled markers shall provide a means of aligning the coder output pulses with respect to their ideal position and for calibration of externally derived pulse trains.

3.7.3.5.1 Marker selection. A crystal marker selector switch shall be located on the front panel of the unit, which will allow the selection of markers having a spacing of 0.1, 1.0, 1.45, 5, or 50 μsec, or a combination of 0.1 and 1 μsec markers. Marker widths shall not exceed the following:

Markers (μsec)	Width (μsec)
0.1	0.05
1.0	0.05
1.45	0.065
5.0	0.05
50.1	1.0

3.7.3.5.2 Marker display. The output of the crystal marker generator shall be in a form suitable for amplitude modulation of the CRT trace, consisting of sharp indices or "pips" superimposed upon a "DC" level. The marker and level amplitudes shall be separately variable (see 3.7.3.5.4). The deflection shall be positive. When used together, the 0.1 and 1.0 μsec markers shall be readily distinguishable from each other.

3.7.3.5.3 Marker trigger. Channel No. 2 shall automatically trigger internally when crystal markers are selected.

3.7.3.5.4 A panel control concentric with the marker selection shall control the marker level (0.1, 1, 1.45, 5, and 50 μsec marks) over a range of less than 0.5 to at least 2 cm. An off position shall be provided by the marker selector.

3.7.3.5.5 Marker spacing accuracy. The markers shall have an accuracy of plus or minus 0.02 percent.

3.7.3.5.6 Marker phasing. Phasing of display video with the markers shall be accomplished using the delay sweep capabilities of the oscilloscope.

3.7.3.6 Sweep generator unit. The sweep generator unit shall perform the following functions:

3.7.3.6.1 Sweep duration. The following calibrated sweep durations shall be available with the controls located on the front panel of the unit when triggered at the repetition rates specified in 3.7.2.6. All sweeps shall be of sufficient amplitude to fill the mask opening. Sweep speed accuracy shall be ± 5 percent.

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0.1 μsec/cm	0.1 msec/cm
0.2 μsec/cm	0.2 msec/cm
0.5 μsec/cm	0.5 msec/cm
1.0 μsec/cm	1.0 msec/cm
2.0 μsec/cm	2.0 msec/cm
5.0 μsec/cm	
10 μsec/cm	
20 μsec/cm	
50 μsec/cm	

3.7.3.6.2 Sweep direction. The horizontal sweep shall be from left to right when facing the CRT.

3.7.3.6.3 Sweep trigger and repetition. The horizontal sweep shall be triggered by either of the triggers specified in 3.7.2.6.1, at the specified trigger repetition rates. The sweep plus sweep trigger delay recovery times shall not exceed 200 μsec under all combinations of trigger repetition rates and delays.

3.7.3.6.4 Sweep linearity. The 10 percent portion of the physical length of the sweep, beginning 10 and 80 percent, respectively, from the start of the sweep, shall not differ in time duration from that portion of its physical length lying between 45 and 55 percent from the start of the sweep, by more than plus or minus 5 percent.

3.7.3.6.5 Strobe selectors. The strobe selectors shall be self-restoring following release of the strobe function selector. The selectors shall be located on the front panel of the unit and perform the following functions:

a. Actuating the sweep strobe will return the sweep start to the zero reference and cause a sample of the selected delayed sweep trigger to be displayed on the CRT as a positive deflection.
b. Actuating the trigger strobe will return the sweep start to the zero reference and cause a sample of the delayed trigger output to be displayed as a negative deflection on the CRT. (This provides an accurate indication of the trigger delay with respect to the zero reference.)

3.7.3.7 Display unit.

3.7.3.7.1 Cathode ray tube (CRT). A 5-inch diagonal flat faced CRT shall be employed using "A" type presentation. A scale, internal to the CRT shall be provided, located at the vertical center line, graduated in fifths of a centimeter, to measure the vertical displacement of the beam. The vertical scales shall extend 3 cm above and below the center of the tube face. A similar scale shall be provided for horizontal measurements, and shall be located approximately on the horizontal center line of the tube face. The horizontal scale shall extend 5 cm on each side of the vertical center.

3.7.3.7.2 Focus. A focus control shall be provided on the front panel to adjust the focus of the trace on the CRT screen.

3.7.3.7.3 Intensity. A control shall be located on the front panel for controlling the intensity of the beam. The intensity control shall have sufficient range to vary the intensity of the beam from high to complete cut-off.

3.7.3.7.3.1 Astigmatism. An astigmatism control shall be provided on the front panel to adjust the astigmatism of the trace on the CRT screen.

3.7.3.7.4 Horizontal and Vertical Centering. Control knobs shall be located on the front panel for controlling the following functions:

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a. Horizontal Centering. The horizontal control shall produce a range of control sufficient to center the trace on the CRT screen. This control shall move the start or finish of the trace to within 2 cm of the center of the screen.
b. Channel No. 1 Vertical Centering. The Channel No. 1 vertical centering control shall produce a vertical displacement of the trace of at least plus or minus 3 cm from the center position, for the main video channel.
c. Channel No. 2 vertical centering. The channel No. 2 vertical centering control shall perform the identical function for the No. 2 vertical channel.

3.7.3.7.5 Vertical amplifier. The vertical amplifier for each channel shall have the following characteristics, selected by means of its selector switch on the front panel. (The gain of each vertical amplifier shall be adjusted by means of a convenient calibration control.) Undershoot and overshoot shall be less than 5 percent of the pulse amplitude with a pulse rise time of 0.02 µsec or less.

Sensitivity ±5 percent	Rise Time
0.05 volt per cm	Less than 0.03 µsec
0.1 volt per cm	Less than 0.03 µsec
0.2 volt per cm	Less than 0.03 µsec
0.5 volt per cm	Less than 0.03 µsec
1.0 volt per cm	Less than 0.03 µsec
2.0 volt per cm	Less than 0.03 µsec
5.0 volt per cm	Less than 0.03 µsec
10.0 volt per cm	Less than 0.03 µsec
20.0 volt per cm	Less than 0.03 µsec

3.7.3.7.5.1 Vertical linearity. With the gain of the vertical amplifier adjusted to obtain 2 cm deflection with a voltage of 2 volts peak applied to the video input jack, and with the video sensitivity selector set for normal sensitivity, an input voltage of 1.0 volt shall cause a deflection of 1.0 plus or minus 0.05 cm, and an input voltage of 5.0 volts shall cause a deflection of 5.0 plus or minus 0.25 cm. Base line shift shall be directly proportional to the duty factor of the video signal in the AC coupling mode only. The check shall be made with a 1 µsec pulse of both positive and negative polarity.

3.7.3.7.6 Video input jack. The impedance looking into the video input jack shall be not less than that presented by a network consisting of 750,000 ohms in parallel wiht a capacitance of 45 µµf. Specified probe accuracy shall be obtained with any of the probes meeting the requirements specified herein, when connected to the video input jack. The equipment shall not be damaged by dc voltages of 500 volts applied to the panel connector when the 75-ohm termination specified in 3.7.3.7.7 is not switched in parallel with the input.

3.7.3.7.7 Switches on the front panel shall connect a 75-ohm ±5 percent 2-watt resistor in parallel with each video input jack.

3.7.3.7.8 Switches on the front panel shall provide for an AC or a DC coupled display of the video input.

3.7.3.7.9 Polarity. Positive input pulse applied to the video input jack shall cause upward deflection of the spot on the oscilloscope screen. Negative input pulses shall cause downward deflections of the spot.

3.7.3.7.10 Rise time. The rise time of a video pulse viewed at the output of the amplifier shall be not greater than that given by the following equation:

$$t_{out} = \sqrt{t^2_{in} + 0.0009}$$

where t out = rise time of output pulse

where t in = rise time of input pulse

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3.7.3.7.11 Decay time. The decay time of a video pulse viewed at the output of the amplifier shall be not greater than that given by the following equation:

$$t_1 \text{ out} = \sqrt{t_1{}^2 \text{ in} + 0.0009}$$

where t_1 out = decay time of output pulse

where t_1 in = decay time of input pulse

3.7.3.7.12 Frequency response. The frequency response of the vertical amplifier shall be essentially flat (within plus or minus 1.5 decibels (dB) from dc to 8 MHz. The rolloff of frequency response shall not exceed 3 dB at a bandwidth of 12 MHz. The phase characteristics shall be adequate to insure that the slope of a 5000 μsec pulse will be increased by not more than 10 percent. The higher frequency response and phase characteristics shall be adequate to insure the rise and decay characteristics specified in 3.7.3.7.5, 3.7.3.7.10 and 3.7.3.7.11.

3.7.3.7.13 Frequency mark input. A separate jack, type UG-625B/U in accordance with MIL-C-39012/21 or equivalent shall be provided on the front panel of the module to serve as input of the sweep frequency marker pulses of 3.7.4.3.4 and 3.7.5.3.4. The input shall cause these frequency marker pulses to add algebraically to the main video channel. When connected to the swept frequency marker pulses of 3.7.4.3.4 or 3.7.5.3.4, vertical deflection shall be between 1 and 2 cm, negative.

3.7.3.8 Video calibrator. The video calibrator shall supply a square wave as specified herein for calibration output jack type UG-625B/U or equivalent on the front panel.

3.7.3.8.1 Output. The output of this unit shall be a 0.2, 0.5, 1, 2, 5, 10 or 20 volts peak to peak calibration square wave at a PRF of 1000 ± 20 percent Hz. The rise and decay time of the calibration square wave shall be equal to or less than the following with an external load of 500K ohms and the specified capacity:

Calibration Pulse Volts	T_R μsec	T_D μsec	C_L μsec
0.2	0.25	1	40
0.5	0.25	1	40
1	0.25	1	40
2	0.25	1	40
5	0.25	1	40
10	0.25	1	40
20	0.25	1	20

During the decay time the square wave shall have a negative overshoot of not more than 10 percent of the maximum positive amplitude. The ac output impedance of the video calibrator shall not exceed 2,100 ohms. The amplitudes of the output levels shall be accurate to ± 3 percent.

3.7.4 Interrogator module.

3.7.4.1 General design. Basic design of the Interrogator Module shall permit removal of the unit from the equipment case and allow operation separately from the rest of the test set for maintenance purposes. Figure 4 is the approved front panel design for the Interrogator Module. Deviation from this configuration requires approval by the command or agency concerned.

3.7.4.2 Circuit elements. The equipment functions shall be provided by four (4) functional areas.

(a) Transmitter
(b) 60 MHz sweep generator
(c) Metering circuitry
(d) System simulator

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3.7.4.3 Transmitter.

3.7.4.3.1 Frequency. The transmitter output frequency shall be crystal controlled and shall operate at a frequency of 1090 MHz ±0.01 percent except in the "sweep" mode.

3.7.4.3.2 Power. The transmitter output power shall be sufficient to insure operation in conjunction with the RF module. The intent is that this signal shall be supplied to the RF module as specified in 3.7.5.4. The output at the RF module SIG GEN jack shall then be adjustable over the range from 0 dBm to -100 dBm and shall have a stability of ±1 dB when set to any value within this range.

3.7.4.3.3 RF Sweep Generator. The transmitter shall be provided with an RF frequency sweep capability. When in the XTAL mode, the transmitter shall operate at a fixed crystal controlled frequency of 1090 MHz. When in the SWEEP mode, the transmitter shall generate a swept CW signal at a 250 Hz ±5 percent rate over the range of 1070 to 1110 MHz. A front panel switch shall select the operating mode.

3.7.4.3.4 Sweep frequency markers. A frequency marker system shall be provided consisting of crystal markers resonating in time synchronism with the RF sweep frequency. The markers shall provide a mark for the center frequency, 1090 MHz, and additional marks at ±3, ±5, ±10, and ±20 MHz. The markers shall be provided on a separate front panel output jack for presentation on the oscilloscope. Frequency marker accuracy shall be within ±0.1 MHz. Markers shall be available on a jack type UG-625B/U in accordance with MIL-C-29012/21 or equivalent.

The markers signals shall have the following characteristics when terminated by 10 K ohms:

Amplitude:	5 ±1.0 V negative
Pulse Width:	2 ±0.5 μsec

3.7.4.3.5 Sweep sync output. A signal shall be provided that is in time synchronism with the start of the frequency sweep. The signal shall be capable of externally triggering the oscilloscope module. The sync signal shall be available on a front panel jack type UG-625B/U in accordance with MIL-C-39012/21 or equivalent.

The signal shall have the following characteristics when terminated by 10 K ohms:

Amplitude:	8 ±2 volts negative
Pulse Width:	400 ±40 μsec

3.7.4.4 60 MHz sweep generator. The module shall contain a sweep generator for producing a swept intermediate frequency (IF) 60 MHz output signal. The sweep generator shall accept the fixed 1030 MHz RF output from the RF module, and shall apply this to the swept 1070 to 1110 MHz RF signal generated internally to produce an output IF signal sweeping over the range of 40 to 80 MHz.

3.7.4.4.1 Power output. The swept output signal shall be adjustable over the range of -20 dBm to -80 dBm. Power output adjustment shall be accomplished by means of a control on the Interrogator module. Power output accuracy shall be within ±3 dB for all settings of the attenuator. A calibration control may be provided to preset a reference power level utilizing the front panel meter.

3.7.4.4.2 IF output stability. The swept IF output power variation shall not be greater than ±0.5 dB over the full sweep range.

3.7.4.4.3 Output impedance. The swept IF output shall be designed to operate into a 50 ohm nominal load.

3.7.4.4.4 Sync and markers. The sweep sync and frequency markers generated by the Interrogator Module in the SWEEP mode shall also function in conjunction with the 60 MHz sweep generator to provide a sweep start signal and crystal frequency markers (see 3.7.4.3.4 and 3.7.4.3.5).

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3.7.4.4.5 Modulation. It shall be possible to modulate the swept output signal by modulating the fixed 1030 MHz applied signal. This shall be accomplished through a jack on the RF Module.

3.7.4.4.6 Interface. The swept 40 - 80 MHz IF output shall be available through a front panel jack type UG909 in accordance with MIL-C-39012/19 or equivalent.

3.7.4.5 Pulse counter. A pulse counter shall count the repetition frequency of the external trigger (see 3.7.2.6.3) the demodulated output (see 3.7.5.8.3), or the pulse trains (see 3.7.2.4 and 3.7.2.5.3.b). The external trigger shall be provided to the pulse counter through a front panel jack type UG-625B/U in accordance with MIL-C-39012/21 or equivalent.

3.7.4.5.1 Pulse counter ranges. There shall be two pulse counter ranges: 0 to 1000, and 0 to 10,000 pps.

3.7.4.5.2 Pulse counter accuracy. The accuracy of the pulse counter shall be within plus or minus 5 percent at full scale and within plus or minus 10 percent at 1/10 full scale.

3.7.4.5.3 Pulse counter indication shall be on a meter.

3.7.4.5.4 Meter switch. A control shall be located on the front panel and so connected as to indicate:

a. 0 to 1000 scale on pulse counter.
b. 0 to 10,000 scale on pulse counter.
c. Calibration level for 60 MHz signals (if required).

3.7.4.6 Radar system simulators. The module shall generate a series of video signals to simulate the operation of a Radar/IFF System for the benefit of automatic IFF data systems. Simulated video signals generated shall include an azimuth gate, active readout gates, and simulated radar. An internal cable from the SIF Module shall provide the trigger (see 3.7.2.6.2) for the active readout gates and radar simulators.

3.7.4.6.1 Azimuth signal generator. The equipment shall generate a simulated azimuth gate representing a sector of sweep by an antenna. The equipment shall in addition generate a simulated North pulse and simulated azimuth change pulses having angular relationships to the azimuth gating.

3.7.4.6.2 Azimuth change and North. The equipment shall generate a series of azimuth change pulses representing the simulated output of a 4096 bit digital encoder. The pulse rate shall simulate the antenna rotation of 6 rpm ±10 percent. At each 4096th change pulse, a separate simulated North pulse shall be generated.

3.7.4.6.3 Output pulse characteristics. Azimuth change and North pulse output characteristics shall be as follows when loaded by 75 ohms ±5 percent.

Polarity	Positive
Amplitude	3 ±0.5 V
Pulse width	4 ±0.5 μsec
Rise time	0.1 μsec max.
Fall time	0.2 μsec max.

3.7.4.6.4 External North. The front panel switch shall be provided to accept an extennal North pulse having the following characteristics:

Polarity	Positive
Amplitude	4 to 5 volts
Pulse width	0.5 to 20 μsec
Rise time	0.2 μsec max.

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In the external North position the switch shall cause the internally generated North pulse to be disabled, and shall cause the azimuth change pulse generator to reset to the reference or North condition.

3.7.4.6.5 Panel jacks. The azimuth change, North out and EXT North shall be interfaced on front panel jacks type UG-625B/U in accordance with MIL-C-39012/21 or equivalent.

3.7.4.6.6 Azimuth gate. A gating signal shall be generated simulating a sector of sweep of the simulated antenna rotation. An internal control shall permit adjustment of the sector angle from 3° to 10° in width, based on the simulated antenna rotation rate of 6 rpm (36° per second). An external control shall be provided for moving the start of the sector angle from a point less than 45° beyond North to a point greater than 135° beyond North. The azimuth gate shall internally control the active readout signals.

3.7.4.6.7 Active readout generator. The equipment shall contain circuits for generating active readout gates simulating range limits, which in conjunction with the azimuth gate produce simulated target areas bound by range and azimuth limits. The following signals shall be generated.

a. X readout gate. The X readout gate shall be generated 170 ±20 μsec following the input trigger. The X readout gate shall be 240 μsec in duration.
b. Y readout gate. The Y readout gate shall be timed so as to fall completely within the X readout gate. The Y readout gate shall occur within 30 ±10 μsec of the start of the X readout gate and shall be 180 μsec in duration.
c. Positive readout gate. The positive readout gate shall be a positive pulse coincident with the leading edge of the Y readout gate.
d. Negative readout gate. The negative readout gate shall be a negative gate coincident with the entire Y readout gate.
e. SIF target gate. The SIF target gate shall be a positive gate, coincident with the Y readout gate and capable of inhibiting the SIF reply train in the SIF Generator Module at all times except during the gate duration. (see 3.7.2.5.4)

3.7.4.6.7.1 Pulse characteristics. The readout gate characteristics shall be as follows when terminated by 75 ohm ±5 percent.

	X	Y	Pos	Neg
Polarity	+	+	+	-
Amplitude	2 ±0.5V	2 ±0.5V	20V ±2V	4.8 ±0.2V
Pulse Width	240 ±30 μsec	180 ±20 μsec	1 ±0.25 μsec	180 ±20 μsec

3.7.4.6.7.2 Output jacks. The X, Y, positive, negative and SIF target signals shall be available on front panel jacks type UG-625B/U in accordance with MIL-C-39012/21 or equivalent.

3.7.4.6.8 Radar simulator and noise generator. The equipment shall provide a simulated radar signal in the presence of noise simulating actual response of a radar system video detector.

3.7.4.6.9 Simulator. The radar simulator shall generate a series of pulses representing three targets. The pulse pattern shall be initiated by the input trigger. The first pulse shall have a duration of 300 ±30 μsec. The second pulse shall follow the first pulse by 1 ±0.5 μsec and shall have a duration of 1 ±0.2 μsec. The third pulse shall follow the second by 1 ±0.5 μsec and shall have a duration of 5 ±1 μsec.

3.7.4.6.10 Pulse amplitudes. The three radar pulses shall have a maximum amplitude of 3 volts, 1 volt and 2 volts ±10 percent respectively when terminated by 75 ±5 percent ohms. A control shall be provided on the front panel to vary the amplitude of the 3 volt pulse from maximum to less than 0.1 volt. The three pulses shall maintain a 3:1:2 voltage ratio at any setting of the control.

3.7.4.6.11 Noise generator. The equipment shall provide a noise generator generating unlimited "white" noise of positive polarity in the bandwidth between 3 MHz and 10 MHz. The noise generator shall have a

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maximum amplitude of at least 1.5 volts, rms, when terminated by 75 ohms. A front panel control shall permit adjustment of the noise level from maximum to less than 0.1 volt rms.

3.7.4.6.12 Output jack. The simulated radar and noise signals shall be made available as mixed signals at a single front panel jack type UG-625B/U in accordance with MIL-C-39012/21 or equivalent.

3.7.5 RF Module.

3.7.5.1 General design. The RF Module shall have complete test capabilities at the center frequency of 1030 MHz and shall have provisions for operation at 1090 MHz in conjunction with the Interrogator module. Basic design of the RF module shall permit removal of the unit from the equipment case and allow for operation separately from the rest of the test set for maintenance purposes. Figure 5 is the approved front panel layout of the module. Deviations from this configuration require approval of the command or agency concerned.

3.7.5.2 Circuit elements. The equipment function shall be provided by four functional areas.

a. Transmitter
b. Modulators
c. Attenuators
d. Demodulators

3.7.5.3 Transmitter characteristics.

3.7.5.3.1 Frequency. The transmitter output frequency shall be crystal controlled and shall operate at a frequency of 1030 MHz ±0.01 percent except in the sweep mode.

3.7.5.3.2 Power output. The transmitter power output level at the test set interface jack shall be adjustable over the range of 0 dbm to -100 dBm and shall have a stability of ±1 dB when set to any value within this range. The test set shall be capable of delivering the specified power while operating at 100 percent duty cycle.

3.7.5.3.3 RF sweep generator. The RF Module transmitter shall be provided with an RF frequency sweep capability. When in the XTAL mode, the module transmitter shall operate at the fixed crystal controlled frequency of 1030 MHz. When in the sweep mode, the module transmitter shall generate a swept CW signal at a 250 Hz ±5 percent rate over the range of 1010 MHz to 1050 MHz. A front panel switch shall select the operating mode.

3.7.5.3.4 Sweep frequency markers. A frequency marker system shall be provided consisting of crystal markers resonating in time synchronism with the RF sweep frequency. The markers shall provide a mark for the center frequency, 1030 MHz, and additional marks at ±3 MHz, ±5 MHz, ±10 MHz and ±20 MHz. The markers shall be provided on a separate front panel output for presentation on the oscilloscope. Marker accuracy shall be better than ±0.1 MHz. Markers shall be available on a connector type UG-625/U in accordance with MIL-C-39012/21 or equivalent.

The signals shall have the following characteristics when terminated by 10 K ohms:

Amplitude	5 ±1.0 volts negative
Pulse Width	2 ±0.5 μsec

3.7.5.3.5 Sweep sync output. A signal shall be provided that is in time synchronism with the start of the frequency sweep. The signal shall be capable of externally triggering the oscilloscope module. The sync signal shall be available on a front panel jack type UG-625/U in accordance with MIL-C-39012/21 or equivalent.

The signal shall have the following characteristics when terminated by 10 K ohms:

Amplitude	8 ±2 volts negative
Pulse Width	400 ±40 μsec

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3.7.5.4 Alternate CW Input. The module shall be provided with an alternate CW input jack to accept an RF signal 1090 MHz in place of the internal 1030 MHz transmitter. An RF switch shall be provided to disconnect the internal transmitter and insert the external signal upon activation of a front panel control. It is intended that 1090 MHz signal be supplied by the Interrogator Module. The front panel jack shall be type UG-909/U in accordance with MIL-C-39012/19 or equivalent.

3.7.5.5 Modulation. The signal generator shall be capable of being satisfactorily pulse modulated with pulse trains as specified in 3.7.2.4, 3.7.2.5, or any pulses corresponding to those in 3.7.2.7.4.1 through 3.7.2.7.4.5 and 3.7.2.7.6. The RF signal generator path shall be split into two parallel legs, each leg provided with a separate modulation capability. The modulated paths shall then be recombined into a single generator path.

3.7.5.5.1 Delay. The delay between the modulator pulse applied to the signal generator and the signal generator output pulse shall not exceed 0.3 μsec.

3.7.5.5.2 Modulation fidelity. When modulated by SIF pulse trains of nominal characteristics, the rf signal shall have the following characteristics:

(a) Rise time 0.05 - 0.1 μsec.
(b) Decay time 0.05 - 0.15 μsec.
(c) Pulse train amplitude variations as specified in 3.7.2.4.5.
(d) The change in pulse width (modulator input vs. rf output) shall be less than plus or minus 0.025 μsec, and provisions for monitoring the rf output provided.
(e) The change in SIF Code pulse spacing (due to changes of pulse width) shall be less than 0.01 μsec with any 2 pulses as reference.

3.7.5.5.3 When modulated by the SIF Module challenge pulses, the rf signal shall have the following characteristics:

(a) Rise time 0.05 to 0.1 μsec.
(b) Decay time 0.05 to 0.15 μsec.
(c) The change in pulse width (modulator input vs. rf output) shall be less than 0.025 μsec.
(d) The spacing between the rf output pulses shall correspond to the modulation pulses within ±0.01 μsec.

3.7.5.5.4 Main signal generator leg. The main modulator shall have an on-to-off ratio of 60 db or better A front panel jack shall be provided to accept external modulation signals. The modulation input signal shall be 0 ± 0.5 volt for the fully "off" condition and 4.0 ± 0.5 volts for the fully "on" condition. The equipment shall not be damaged by modulation signals up to 28 vdc. The front panel jack shall be type UG-625B/U in accordance with MIL-C-39012/21 or equivalent.

3.7.5.5.5 Auxiliary signal generator leg. The auxiliary modulator shall be indentical to the main modulator and shall be provided with a separate identical front panel modulation input jack. In addition, the auxiliary modulation leg shall be provided with a series attenuator to provide up to 60 dB of attenuation with respect to the main signal generator leg. With the auxiliary attenuator set for minimum attenuation, the auxiliary signal generator leg power level shall be 3 dB higher than the main signal generator leg.

3.7.5.5.6 Modulation override (CW). A front panel jack shall supply a fixed positive DC voltage such that if either modulation video input is patched to this jack, that modulator shall be held in the fully on condition for continuous wave RF operation. The jack shall be type UG 625B/U in accordance with MIL-C-39012/21 or equivalent.

3.7.5.6 Auxiliary attenuator control. A front panel control identified by the work "Aux Attenuator" shall be provided for the attenuator. It shall have an indicator calibrated in dB. The indicator shall have a range

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of at least 60 dB divisions. The rf output shall be within plus or minus 3 dB of the indicator setting when the attenuator is set to the 0 dB indicator setting, over the entire range of frequencies and pulse modulation. Between the frequencies of 1010 and 1110 MHz, the accuracy shall be within plus or minus 1.5 dB. The differential attenuation between any two consecutive 10 dB divisions shall be 10 ± 0.5 dB.

3.7.5.7 Output attenuator. An attenuator shall be provided, capable of varying the combined modulated signal generator output during the pulse continuously between 0 and -100 dB below 1 milliwatt, when matched to a 50-ohm line. The output impedance of the attenuator shall be such that when terminating a 50-ohm line, it shall not produce a standing wave ratio of more than 1.8 from 0 to 10 dB attenuator setting, nor more than 1.4 for attenuator settings above 10 dB.

3.7.5.7.1 Output attenuator control. A front panel control identified by the word "Output Attenuator" shall be provided for the attenuator. It shall have an indicator calibrated in dB below 1 milliwatt under matched load conditions (50-ohm). The indicator shall have a range of at least 100 dB divisions. The rf output shall be within plus or minus 3 dB of the indicator setting when the attenuator is set to the 0 dB indicator setting, over the entire range of frequencies and pulse modulation. Between frequencies of 1010 and 1110 MHz, the accuracy shall be within plus or minus 1.5 dB. At the frequency of 1030 MHz, the accuracy shall be within plus or minus 1.0 dB. The differential attenuation between any two consecutive 10 dB divisions shall be 10 ± 0.5 dB.

3.7.5.7.2 Signal generator input-output connector. The output of the signal generator from the attenuator shall be available at a BNC connector on the front panel, identified by "SIG GEN in-out". The jack shall be type UG-909/U or equivalent. Directional elements shall be incorporated within the test set such that the test set RF input signal shall also be accepted through the same jack.

3.7.5.7.2.1 Output level stability. In the sweep mode, power level variations shall be less than ± 0.5 dB.

3.7.5.8 Demodulation. The demodulation section shall consist of a high power input coupler, video detector and frequency probe circuits.

3.7.5.8.1 High power input coupler. A front panel RF connector shall be provided to accept RF signals up to 15 KW peak power, (25 watts average). The RF input shall present a load of 50 ± 5 percent ohms and a maximum VSWR of 2 dB. The front panel jack shall be a type "C" female connector UG-568/U in accordance with MIL-C-39012/12 equivalent.

3.7.5.8.2 Power probes. Two probes shall be provided in the input coupler of 3.7.5.8.1. The probes shall be sufficiently isolated from the high power input to maintain a signal level within the range of 0.5 to 50 watts which can be accepted by the RF module signal generator input/output jack. The first probe, for high level signals shall sufficiently attenuate RF input signals in the range of 500 watts to 15 KW peak. The second probe, for low level signals shall sufficiently attenuate RF input signals in the range of 50 to 2000 watts. The appropriate probe shall be capable of being externally patched into the module Sig-Gen In/Out jack. The front panel probe jacks shall be type UG-909/U or equivalent.

3.7.5.8.3 Video output. The module shall be provided with a demodulator to accurately reproduce the pulse shape and accurately represent the pulse power of the RF input signal. A peak pulse power input of 0.5 watt shall produce a video output of 0.5 volt minimum when terminated by 75 ohms. The video output shall be valid with a peak power input to the Sig-Gen In/Out terminal of 50 watts with an accuracy within ± 1 dB over the frequency range.

3.7.5.8.4 Calibration. Calibration facilities shall be provided with each equipment to permit power determination in decibels referenced to 1 watt (dbw) to an accuracy of ± 1 dB.

3.7.5.8.5 Pulse video fidelity. At the same time that pulse power is indicated, pulse shape of the RF signal video shall be preserved. At the highest power level, distortion of pulse width shall remain within ± 20 nanosec and distortion of pulse rise and fall time shall remain within ± 10 nanosec, when terminated by 75 ohms. The video output shall be provided on a front panel jack type UG-625B/U in accordance with MIL-C- 39012/21 or equivalent.

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3.7.5.8.6 Frequency probe video. The module shall be provided with a frequency determining circuit to accurately provide a tuning indication for an input RF pulse which is 60 MHz away from the swept RF generator frequency. The circuit shall operate with either the swept internal RF transmitter frequency (1010-1050 MHz) or the external swept cw input frequency (1070-1110 MHz). The tuning indication shall be in the form of a video pulse suitable for presentation on the oscilloscope. The pulse shall have a minimum amplitude of 0.05 volt for an input RF peak power level of 0.5 watt, when terminated by 75 ohms. When used in conjunction with the Oscilloscope module the equipment shall be capable of measuring RF signals near 1030 or 1090 MHz to an accuracy of ± 0.2 MHz. The frequency probe video shall be provided on a front panel jack type UG-625B/U in accordance with MIL-C-39012/21 or equivalent.

3.7.5.9 Receiver side lobe suppression. The module shall be provided with a separate RF output signal at the same frequency as the transmitter (3.7.5.3) or the alternate cw input (3.7.5.4). The output power level shall be adjustable over the range of -17 dBm to -77 dBm by the auxilliary attenuator control (3.7.5.6). It is intended that the auxillary modulator (3.7.5.5.5) be used to modulate the cw signals. The main signal generator path will provide a second source with the output level variable from 20 dB above the selected RSLS output to -95 dBm. The front panel jack shall be type UG-909/U or equivalent.

3.7.6 Accessories. Accessories and an accessory box shall be provided as specified hereinafter.

3.7.6.1 Accessory box. The equipment shall be provided with an accessory box which is part of the equipment case. Provisions shall be made for storage of the following:

(a) Cables
(b) Adapters and probes
(c) Calibration Charts (if required)
(d) Operators Manuals
(e) Spare Bulbs
(f) Extender board(s)

3.7.6.2 Video test probe. Three video test probes shall be supplied. The characteristics of the video probes shall be as follows:

Attenuation Ratios	Input Capacity	Dielectric Volts DC	Dielectric Volts AC Peak to Peak
1:1	50 μμf or less	600	600
10:1	10 μμf or less	600	600
100:1	5 μμf or less	1500	4200

The general assembly of the probe shall consist of a length of shielded flexible cable (5.5 to 6 feet long) terminated by a connector of the BNC series and the probe body. Each probe shall be suitably marked for identification. Design of the probe shall be subject to approval of the command or agency concerned.

3.7.6.3 Radio Frequency Cables.

3.7.6.3.1 General Service Cables Two 5-foot lengths of RF cable, type RG223/U in accordance with MIL-C-17/84, having a UG-88E/U plug in accordance with MIL-C-39012/16 equivalent on each end shall be provided. These cables shall be marked with their attenuation.

3.7.6.3.2 High Power Cable. One 10-foot length of RF cable, type RG-212/U in accordance with MIL-C-17/73, having a UG-626B/U plug in accordance with MIL-C-39012/6 or equivalent on each end shall be provided. This cable shall be marked to indicate its attenuation.

3.7.6.3.3 Radio Frequency Jumper Cable A length of RF cable RG223/U in accordance with MIL-C-17/84 having sufficient length to connect the "SIG GEN IN/OUT" to the "POWER PROBE" connectors shall be provided.

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3.7.6.4 Video Cables.

3.7.6.4.1 General Service Cables. Four 5-foot lengths of RF cable, type RF-62A/U in accordance with MIL-C-17/30, each having a plug UG-260D/U in accordance with MIL-C-39012/16 or equivalent on each end, shall be provided.

3.7.6.4.2 Video Jumper Cable. One length of RF cable RG62A/U in accordance with MIL-C-17/30 having a UC-260D/U plug in accordance with MIL-C-39012/16 or equivalent on each end shall be supplied. The cable shall be of sufficient length to connect the "Video Out" jack specified in 3.7.2.9 with the modulation input jacks specified in 3.7.5.6.4, 3.7.5.6.5, or the oscilloscope jacks specified in 3.7.2.7.6.

3.7.6.5 Adapters. The following adapters shall be supplied:

Type (UG-)	Applicable Specification, Standard or Drawing	Quantity
273/U BNC to UHF	REC49438	3
274B/U BNC to "T"	MS35173 or equivalent	2
201A/U "N" to BNC	MIL-A-27434/37	2
635/U C to BNC	MIL-A-27434/3	1
636A/U C to BNC	MIL-A-27434/4	4
309/U HN to BNC	REB49276 or equivalent	2
50-ohm termination BNC connector		2

3.7.6.6 Connectors. Unless otherwise specified herein, all external jacks for video pulses and trigger pulses shall be similar to type UG-625B/U in accordance with MIL-C-39012/21 or UG-290A/U in accordance with MIL-C-39012/22. Associated plugs shall be similar to UG-260D/U in accordance with MIL-C-39012/16. Adapters shall be provided as specified in 3.7.6.5. Unless otherwise specified herein, all RF external jacks shall be similar to type UG60E/U of REB49224, UG61E/U of REB49228. UG290C/U in accordance with MIL-C-39012/22; associated plugs shall be type UG59E/U of REB49227 or UG88E/U in accordance with MIL-C-39012/16, or plugs and connectors suitable for RG223/U cable in accordance with MIL-C-17/84.

3.7.6.7 Service test cables. Service test cable(s) shall be supplied to enable powering of the modules of the basic equipment when separated from the basic chassis of the equipment for maintenance purposes.

3.7.6.8 Extender Board(s). For maintenance, one extender board shall be provided for accommodating each style of plug in printed circuit board used in the system. The extender board(s) shall have sufficient length to extend the accommodated circuit board out of its card file position to a position of complete two side accessibility. The extender board shall be keyed to prevent accidental reversal of the extended circuit board.

3.7.7 Input voltage. The equipment shall operate within the requirements of this specification from single phase supplies having an ac line voltage of 115 volts plus or minus 10 percent at any frequency from 45 to 420 Hz. The supply voltage may have a form factor between 1.03 and 1.25, and an amplitude factor between 1.16 and 1.68. Taps shall be included on the power transformers for an adjustment at installation to provide for continuous operation at line voltages of 103.0 or 127.0 plus or minus 10 percent ac.

3.7.8 Elapsed time indicator. A dc elapsed time indicator, type MS17321-1 in accordance with MS17321, shall be incorporated in the test set to record cumulative operating time. The meter shall be operative whenever the main power switch of the test set is turned to the ON position.

3.7.9 Primary Power Cables. The connector on the power input end of the cable shall be a type UP131M in accordance with MIL-C-3767/12. The equipment chassis shall be electrically connected to the ground pin on the connector plug.

3.7.10 Circuit Breakers. Circuit breakers shall conform to MIL-C-39019 and shall be used for the primary input power circuitry.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system. The contractor shall provide and maintain an effective inspection and quality assurance system acceptable to the Government covering the equipment being procured. The quality assurance system shall be in accordance with MIL-Q-9858. A quality program plan, as it would apply to the requirements of this contract, shall be prepared prior to first article inspection (see 6.1.2). If first article inspection is not required, the quality program plan shall be prepared prior to production inspection. This program plan shall include flow charts showing the complete flow of material from inspection on receipt, through all manufacturing processes, to final shipment. The various processes and inspection positions shall be clearly identified on the flow-charts and cross referenced to the detailed inspection procedures and criteria of the written description of the quality control system. Any changes to the quality control system or the quality program plan shall be submitted to the cognizant Government Inspector for approval prior to use.

4.1.2 Government Verification. All quality assurance operations performed by the contractor will be subject to Government Verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, and (b) Government product inspection to measure quality of product to be offered for acceptance. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 Classification of inspection. The methods of examination and testing are as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection
    (1) Production inspection (see 4.4)
    (2) Production control inspection (see 4.5)
    (3) Environmental tests (see 4.6)
(c) Inspection of preparation for delivery (see 4.9)
(d) Government product inspection of delivered product to assure compliance with all requirements of this specification.

4.3 First article inspection. Unless otherwise specified (see 6.1.1) first article inspection shall be conducted on one first article sample and shall consist of the tests specified in MIL-E-16400 except that water cooling and reliability tests are not required. In addition to the preproduction tests specified in MIL-E-16400, the altitude test of MIL-STD-810, Procedure I of Method 500, shall be conducted to verify compliance with 3.5.3.

4.3.1 First article inspection report. A first article inspection report shall be prepared containing all the results of the first article inspection. The report shall include completed Form NAVEXOS 3373 as specified in MIL-S-901 for the shock test and resilient mount data.

4.4 Production inspection. Production inspection shall be made on all equipments in accordance with MIL-E-16400, except that the operating test shall be as specified in 4.4.1.

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4.4.1 Operating test. The equipment shall be completely aligned and calibrated and an operating test shall be conducted for 24 hours. The procedures and reference measurements to be made at the start and completion of the 24-hour test shall be approved by the command or agency concerned.

4.5 Production control inspection. Production control inspection shall consist of the tests specified in MIL-E-16400 except that water cooling tests are not required. Production control inspection shall be conducted on production tested equipment. Reliability testing shall be in accordance with paragraph 4.5.4.

4.5.1 Production control inspection (PCI) tests shall be conducted on a total of four lots of five equipments each during the production life of the contract. The first lot shall be randomly selected from the first month's production with subsequent lots randomly selected at the one-quarter, one-half, and three-quarter points of the production run. Each of the five equipments in each lot selected for PCI tests shall be subjected to all the PCI tests specified in 4.5. Except only one unit from each lot need be subjected to the RFI tests.

4.5.2 If at any time a sample unit fails in any PCI test the contractor shall be required to perform a failure analysis and make corrections as necessary to meet the requirements of this specification. The equipment shall then be retested. It is not required that additional equipments be subjected to PCI tests if there are failures within the PCI sample lot (this does not prohibit the contractor from testing additional equipment at his expense as part of a failure analysis).

4.5.3 If the failure analysis and corrective action discussed in 4.5.2 results in a change of part(s), circuit layout or the mechanical equivalent such action shall be reported to the command or agency concerned immediately. It shall then be determined by the command or agency concerned whether the correction should be made to the equipments already delivered under the contract. No production deliveries shall be made until the initial sample lot has successfully completed the PCI tests unless otherwise specified by the command or agency concerned. For subsequent lots production deliveries shall not be withheld pending completion of PCI tests or pending receipt of government decision on retroactive corrections unless otherwise stipulated by the command or agency concerned.

4.5.4 Reliability testing. The first month's production of test sets (or a maximum of five units therefrom) which shall constitute the lot, shall be evaluated in accordance with the Qualification Phase of Production Reliability tests of MIL-STD-781. After successful qualification, the test set shall be subjected to periodic monthly re-testing with actual test months selected randomly. A total of three separate months out of the remaining production time shall be selected at random (and not in advance) by the cognizant government inspector for demonstration of the reliability of the test set. During each of the months selected, a total of five (5) equipments shall be selected by the government inspector from that month's production and shall be tested in accordance with MIL-STD-781 to demonstrate the required reliability.

4.5.4.1 The reliability tests shall be performed in accordance with Test Level A-1, Test Plan V of MIL-STD-781.

4.5.4.2 Failure. Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification.

4.5.4.3 Rejected equipment. When equipment has been rejected as a result of the reliability test, the whole lot from which it was drawn shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test. In the event that the cause of failure is equipment manufacturer design or part manufacturer designs, corrective action shall be performed on all equipment delivered under the contract.

4.5.4.4 Disposition of tested equipment. Upon successful completion of reliability testing, tested equipment shall be returned to the production area for reinspection and re-conditioning as required and shall be subjected to the production tests of this specification to assure that it meets the requirements of this specification. After reconditioning, reinspection, and retesting, the equipment may be delivered as part of the contract quantity provided it meets all terms and conditions of the contract for acceptance.

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4.5.5 Reinspection of conforming production control test units. Unless otherwise specified, sample units which have been subjected to and passed production control inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.6 Environmental tests. Environmental tests in accordance with MIL-E-16400 are required in the production cycle. At the initial, one-quarter, one-half, and three-quarters points of the production cycle one equipment which has passed the production inspection tests shall be selected by the government inspector and subjected to the environmental tests. The conditions for withholding production deliveries outlined in 4.5.3 for the PCI tests also apply for the environmental tests.

4.6.1 Shock. The test set shall meet the shock requirements of Grade A, Deck Mounted, Class 1, Lightweight and Type A of MIL-S-901. The test set shall not be turned on during the test but shall perform all its functions within the limits specified in this specification at the completion of the test. The method of mounting the equipments for testing shall be subject to the approval of the command or agency concerned.

4.6.1.1 The CRT must be installed during the one-foot hammer drop and must operate within the limits of this specification afterwards. Distortion of the sweep is permissible after the three and five-foot drops or the CRT may be removed for these blows.

4.6.2 Nonconforming environmental sample units. If a sample unit fails the test specified in 4.6, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor can not determine the cause of failure, the matter shall be referred to the contracting officer.

4.6.3 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract, sample units which have been subjected to and passed environmental tests may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage.

4.7 Test procedure list. In order to provide for efficient procedures for accomplishing the testing required by 4.3 through 4.6, the contractor shall be required to submit copies to the command or agency concerned, via the cognizant inspector, (see 6.1.2) prior to approval:

(a) Tests to be performed, including each and every point to be measured.
(b) Test procedure to be followed, including the connection, adjustment, measurement, and disconnection of all the equipment, including the test equipment.
(c) A complete list of the test equipment and accessories (major) to be used, the manufacturer's type number, and the accuracy to be obtained for each and the total accuracy obtainable for the test set-up. Certificates of test equipment calibration accuracy shall be required prior to the actual testing, if the cognizant inspector or the bureau or agency concerned considers any item doubtful.
(d) Block diagrams of the test set-up to supplement (b).
(e) A list of the facilities available at the contractors plant for conducting the first article inspection (service conditions) of MIL-E-16400. If these items are to be subcontracted, the prime contractor shall obtain such a list and forward it to the command or agency concerned.

4.8 Correction of deficiencies found as a result of testing outlined in 4.3 through 4.6 above shall be the responsibility of the contractor. In case of equipments which have been accepted by the Government prior to discovery of the deficiencies, it shall be the responsibility of the contractor to correct the deficiencies by retrofit on approval by the command or agency concerned.

4.9 Inspection of preparation for delivery. Inspection shall be conducted to insure conformance with the requirements of section 5 of this specification.

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 (See 6.1.1).

6. Notes

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

(a) Title, number and date of this specification.
(b) Number of first article samples to be submitted, if other than specified (see 4.3).
(c) Marking, levels of preservation and packaging and packing required (See 5.1)

6.1.2 Contract data requirements. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification includes, but is not restricted to the following:

(a) Reliability program plan (see 3.4.1.1)
(b) Contractors quality assurance system (see 4.1.1)
(c) Test procedure list (see 4.7)

6.2 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - EC

(Project 6625-N448)

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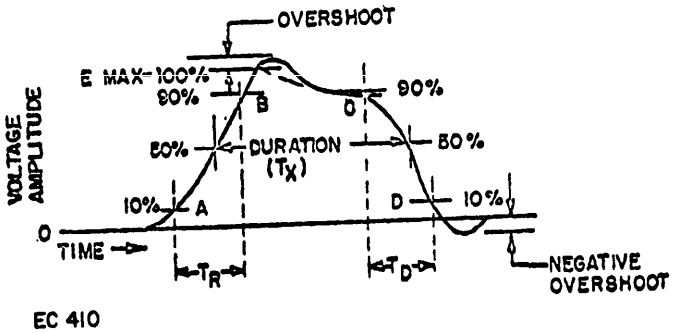


FIGURE 1. Pulse waveform.

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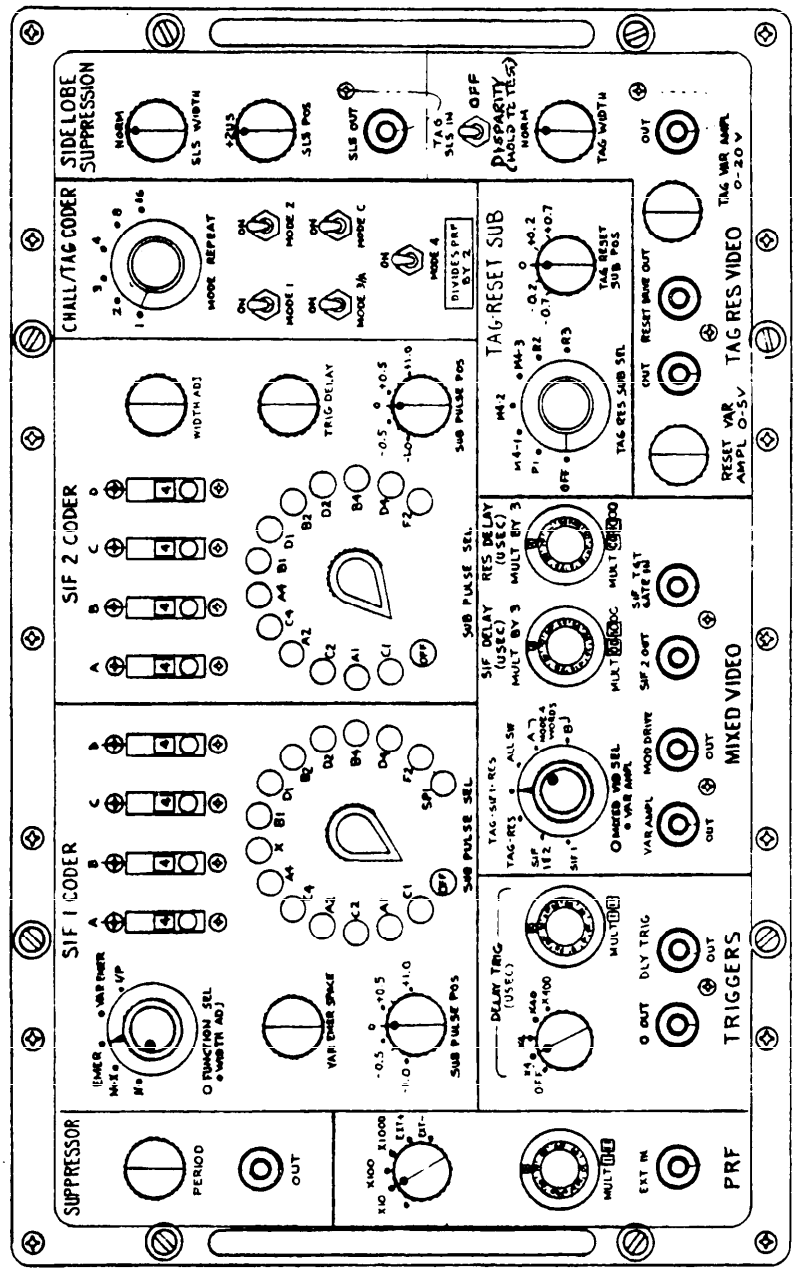


FIGURE 2. SIF module front panel.

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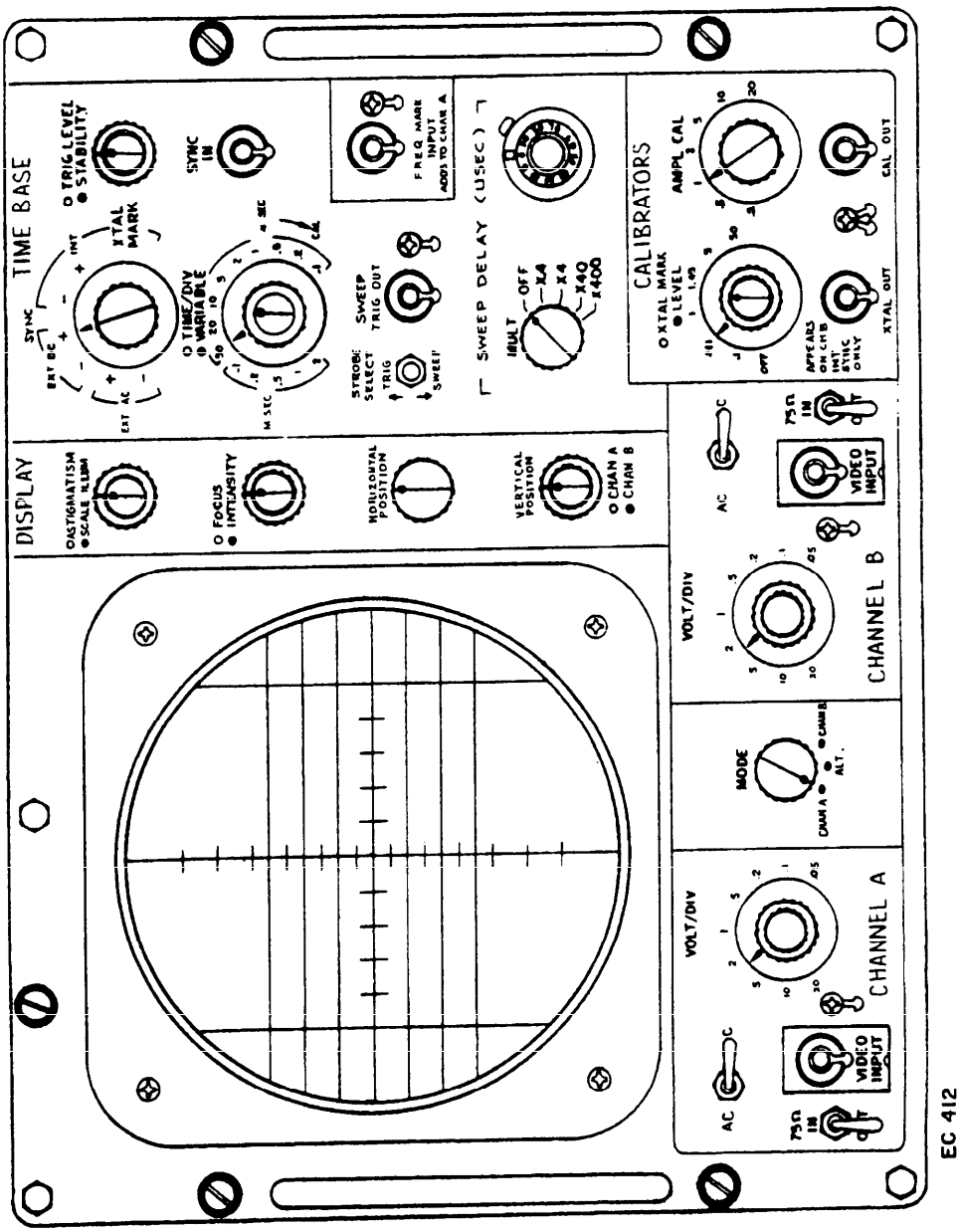


FIGURE 3. Oscilloscope module front panel.

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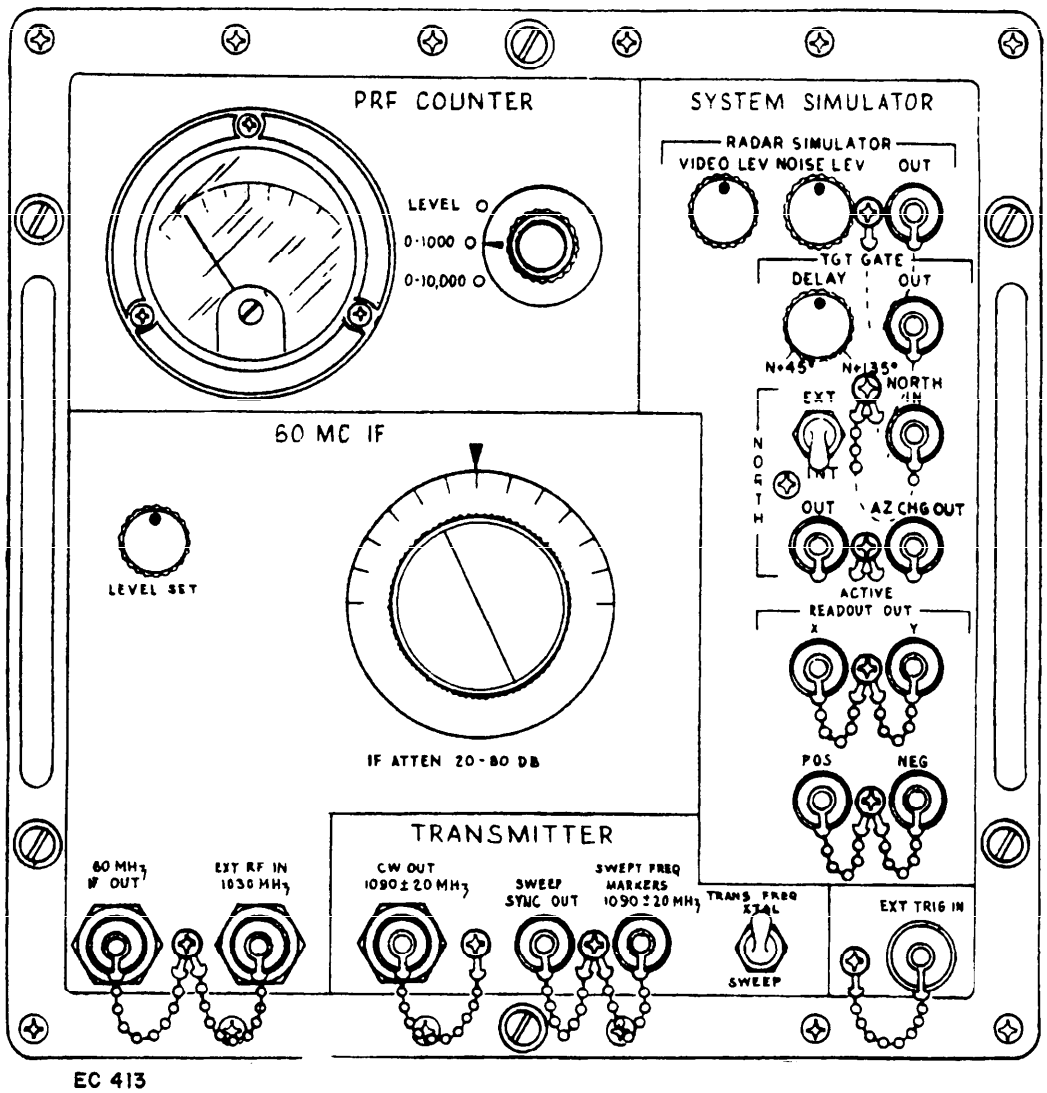


FIGURE 4. Interrogator module front panel.

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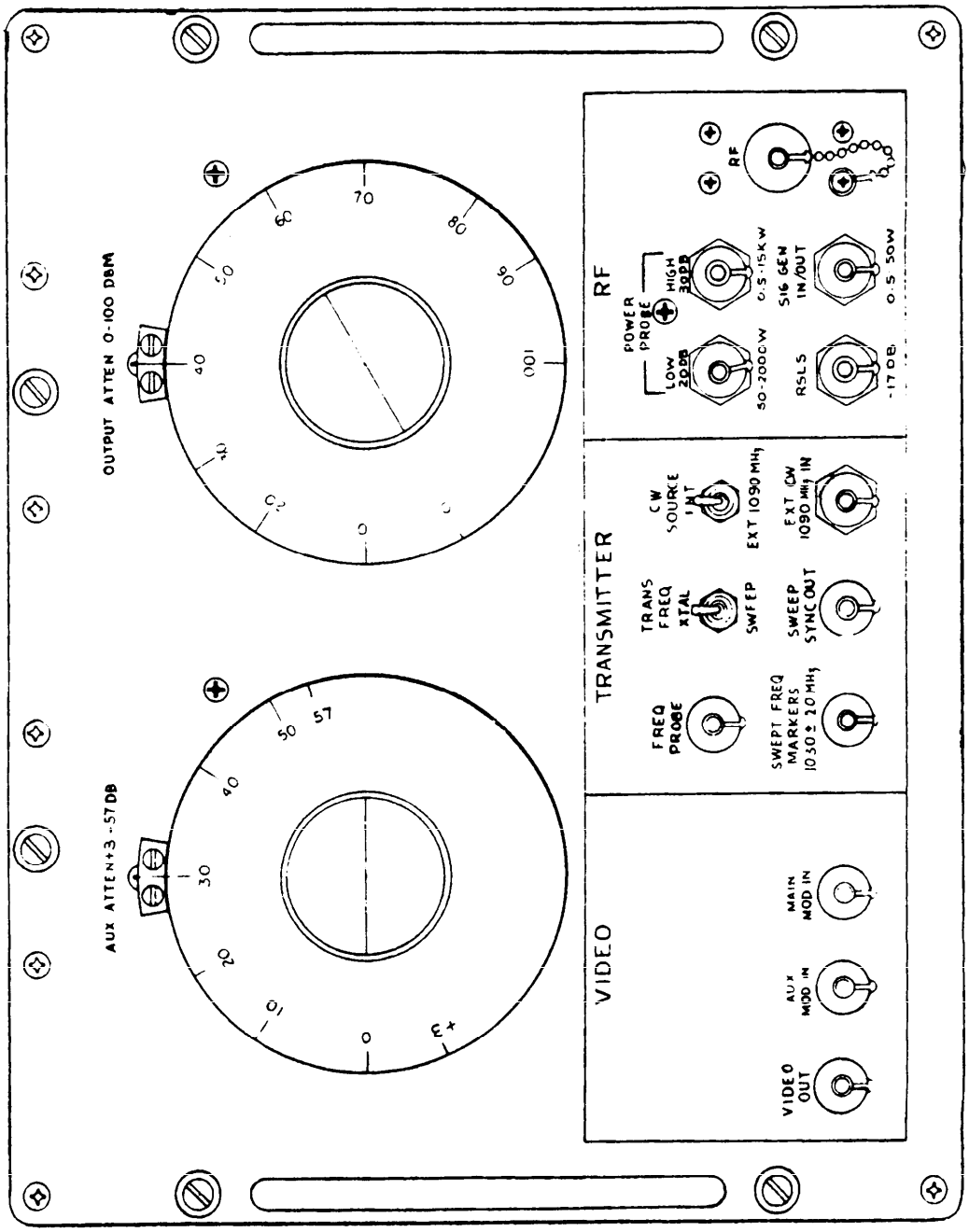


FIGURE 5. RF module front panel.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 4034 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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